Supplementary Material

1 Definitions & Additional Preliminaries

1.1 Round-by-Round (Knowledge) Soundness & Collaborative BCS

In this work, we will prove that our protocols are round-by-round (knowledge) sound [5, 6, 9] is—a strengthening of standard (knowledge) soundness that enables us to prove (knowledge) soundness of an IOP when making it non-interactive using the BCS transformation [3]. More specifically, we will use the notions of round-by-round (knowledge) soundness proposed in [6, 9]. In particular:

Definition 1 (Round-by-Round Soundness)**.** *We say that a k-round IOP(P) for an NP relation $\mathcal{R}$ is round-by-round sound with error ϵ_{rbr} if there exists a doomed set $\mathcal{D}$ of partial and complete transcripts (where a partial transcript is a suffix of $([\pi^{(1)}], r_1, \ldots, [\pi^{(k)}], r_k)$ which is a complete transcript) such that:*

1. *If $\boldsymbol{x} \notin \mathcal{L}_{\mathcal{R}}$, then $(\boldsymbol{x}, \emptyset) \in \mathcal{D}$.*
2. *If $(\boldsymbol{x}, \boldsymbol{tr}) \in \mathcal{D}$ and the transcript $\boldsymbol{tr}$ is complete, the verifier rejects.*
3. *For every $i \in [k]$ and partial transcript $\boldsymbol{tr}_i$, such that $(\boldsymbol{x}, \boldsymbol{tr}_i) \in \mathcal{D}$, then it must hold that $\Pr_{r_{i+1} \xleftarrow{\$} \mathbb{F}}[(\boldsymbol{tr}_i, [\pi^{(i+1)}], r_{i+1}) \notin \mathcal{D}] \leq \epsilon_{rbr}$.*

Definition 2 (Round-by-Round Knowledge Soundness)**.** *We say that a k-round IOP(P) for an NP relation $\mathcal{R}$ is round-by-round knowledge sound with error κ_{rbr} if there exists a PPT extractor $\mathcal{E}$ and a doomed set $\mathcal{D}$ such that:*

1. *$(\boldsymbol{x}, \emptyset) \in \mathcal{D}$ for all possible inputs.*
2. *If for any input $\boldsymbol{x}$ and complete transcript $\boldsymbol{tr}$, holds that $(\boldsymbol{x}, \boldsymbol{t}) \in \mathcal{D}$ then, the verifier rejects.*
3. *For any $i \in [k]$ and partial transcript $\boldsymbol{tr}_i$, such that $(\boldsymbol{x}, \boldsymbol{t}_i) \in \mathcal{D}$, then, for any $\pi^{(i+1)}$ it holds that if: $\Pr_{r_{i+1} \xleftarrow{\$} \mathbb{F}}[(\boldsymbol{t}_i, [\pi^{(i+1)}], r_{i+1}) \notin \mathcal{D}] > \kappa_{rbr}$. Then, the extractor outputs a witness $\boldsymbol{w}$ such that $(\boldsymbol{x}, \boldsymbol{w}) \in \mathcal{R}$.*

Once we proved that our underlying coIOPPs and coPIOPs have a negligible round-by-round (knowledge) soundness error then, we can utilize the following Theorem introduced by Pappas et. al. [12] which enables us to argue about the security of the non-interactive protocols produced by the collaborative BCS transformation:

Theorem 1. *For any coIOPP with round-by-round (knowledge) soundness errors $\epsilon_{rbr}, \kappa_{rbr}$ and any relation $\mathcal{R}$, random oracle $H : \{0,1\}^* \to \{0,1\}^\lambda$ and query bound $Q \in \mathbb{F}$ the coBCS transformation of [12] produces a non-interactive (with respect to the verifier) proof system for $\mathcal{R}$ with:*

- *Adaptive soundness error $\epsilon_{fs} \leq Q\epsilon_{rbr} + 3(Q^2+1)/2^\lambda$ and adaptive knowledge soundness error of $\kappa_{fs} \leq Q\kappa_{rbr} + 3(Q^2+1)/2^\lambda$.*
- *Adaptive soundness error $\epsilon_{qfs} = \Theta(Q\epsilon_{fs})$ and adaptive knowledge soundness error of $\kappa_{qfs} = \Theta(Q\kappa_{fs})$ for Q-$\mathcal{O}(q(\boldsymbol{x})\log l(\boldsymbol{x}))$-query quantum adversaries.*
- *If the IOP is honest-verifier and prover statistical t-zero knowledge, then the non-interactive protocol is statistical t-zero knowledge with error $z^H_{fs} \leq z_{coIOP} + z_{MT}(\lambda, l(|\boldsymbol{x}|), q(|\boldsymbol{x}|), Q) + \frac{Q}{2^{N\lambda/(N-t)}}$ where $z_{MT}(\lambda, l(|\boldsymbol{x}|), q(|\boldsymbol{x}|), Q)$ the blinding error of the Merkle Tree commitment scheme.*
- *If the IOP is honest-verifier and **maliciously secure** t-zero knowledge, then the non-interactive protocol is statistical t-zero knowledge with error $z^M_{fs} \leq z_{coIOP} + z_{MT}(\lambda, l(|\boldsymbol{x}|), q(|\boldsymbol{x}|), Q) + \frac{Q}{2^\lambda}$.*

1.2 Generalized Minimum Distance Decoding for $\mathrm{RS}^{\otimes 2}$

In this section, we show how to adapt the Forney's algorithm to decode RS tensor-codes $\mathrm{RS}[(k,k),(n,n),\mathbb{F}]$ with up to $(n-k)(n-k)/2$ errors. Looking ahead, this algorithm will be used to prove tighter proximity gaps for RS tensor-codes. The adapted algorithm works as follows:

1. For each column $i \in [n]$ of $\mathbf{C}$, find $\mathbf{U}_i \in \mathbb{F}^k$ such that $w_i = \Delta(\mathbf{C}_i, \mathrm{Enc}_{\mathrm{RS}}(\mathbf{U}_i))$ is minimized.
2. For each $i \in [n]$, if $w_i \geq (n-k)/2$, set $\mathbf{U}_i = \perp$. Otherwise, with probability $2w_i/(n-k)$, set $\mathbf{U}_i = \perp$.
3. For every $j \in [k]$, initialize $\mathbf{W}_j = [\mathbf{U}_{i_1}[j], \ldots, \mathbf{U}_{i_s}[j]]$ were $\{i_1, \ldots, i_s\}$ the indexes of the non-erased columns, and use the Berlekamp–Welch algorithm for decoding RS codewords to decode $\mathbf{W}_j$ up to $(n-k-s)/2$ errors. If the algorithm correctly decodes $\mathbf{W}_j$ to $\mathbf{M}_j$ for all $j \in [k]$, it outputs $\mathbf{M} = (\mathbf{M}_1, \ldots, \mathbf{M}_k)$. Otherwise, it repeats step 2.

Correctness. Let $\mathbf{C}^* \in \mathrm{RS}^{\otimes 2}$ be the codeword such that $\Delta(\mathbf{C}^*, \mathbf{C}) < \frac{(n-k)^2}{2}$, e be the number of invalid columns selected by the algorithm at step 2, (e.g., the

columns such that $\mathrm{Enc}_{\mathrm{RS}}(\mathbf{U}_i) \neq \mathbf{C}^*_i$) and s the number of erased columns. To prove the correctness of the algorithm, we will show that (1) for the expected number of e and s it holds that $2e + s < n - k$ and (2) if $2e + s < n - k$, then, the algorithm outputs $\mathbf{M}$ such that $\mathrm{Enc}_{\mathrm{RS}^{\otimes 2}}(\mathbf{M}) = \mathbf{C}^*$.

Proving (1). Let $X_i^{\perp}$ and X_i^e be the random variables such that $X_i^{\perp} = 1$ (and $X_i^e = 1$ resp.) iff $\mathbf{U}_i = \perp$ (and $\mathrm{Enc}_{\mathrm{RS}}(\mathbf{U}_i) \neq \mathbf{C}^*_i$ resp.). We will show that $\mathrm{E}[2X^e + X^{\perp}] = \mathrm{E}[\sum_{i\in[n]}(2X_i^e + X_i^{\perp})] \leq n-k$ which indicates that the expected value of $2e + s$ is smaller than $n - k$. To prove the above we use the exact same argument as in Forney's algorithm for concatenated codes. In particular, for each column $i \in [n]$ and $e_i = \Delta(\mathbf{C}^*, \mathbf{C})$ we consider the following cases:

1. It holds that $\mathrm{Enc}_{\mathrm{RS}}(\mathbf{U}_i) = \mathbf{C}^*_i$. Then, $\mathrm{E}[X_i^e] = 0$ and $\mathrm{E}[X_i^{\perp}] = \frac{2w_i}{n-k} = \frac{2e_i}{n-k}$ where $e_i = \Delta(\mathbf{C}_i, \mathbf{C}^*_i)$. So, $\mathrm{E}[2X_i^e + X_i^{\perp}] \leq \frac{2e_i}{n-k}$.

2. It holds that $\Delta(\mathrm{Enc}_{\mathrm{RS}}(\mathbf{U}_i), \mathbf{C}_i) = w_i \geq \frac{n-k}{2}$. Then, $\mathrm{E}[X_i^e] = 0$ and $\mathrm{E}[X_i^{\perp}] = 1 \leq \frac{2w_i}{n-k} \leq \frac{2e_i}{n-k}$. So, $\mathrm{E}[2X_i^e + X_i^{\perp}] \leq \frac{2e_i}{n-k}$.

3. It holds that $\Delta(\mathrm{Enc}_{\mathrm{RS}}(\mathbf{U}_i), \mathbf{C}_i)) < \frac{n-k}{2}$ but $\Delta(\mathrm{Enc}_{\mathrm{RS}}(\mathbf{U}_i), \mathbf{C}^*_i)) \geq n - k$. In this case, $\mathrm{E}[X_i^{\perp}] = \frac{2w_i}{n-k}$ and $\mathrm{E}[X_i^e] = 1 - \frac{2w_i}{n-k}$ so $\mathrm{E}[2X_i^e + X_i^{\perp}] = 2 - \frac{2w_i}{n-k} = 2\frac{((n-k)-w_i)}{n-k}$. Note that $n - k - w_i \leq \Delta(\mathrm{Enc}_{\mathrm{RS}}(\mathbf{U}_i), \mathbf{C}^*_i) - \Delta(\mathrm{Enc}_{\mathrm{RS}}(\mathbf{U}_i), \mathbf{C}_i) \leq \Delta(\mathbf{C}_i, \mathbf{C}^*_i) = e_i$—where the last inequality steps from the triangular inequality. Hence, $E[2X_i^e + X_i^{\perp}] \leq 2\frac{e_i}{n-k}$

From all the above cases, we have that $\mathrm{E}[X^e + X^{\perp}] \leq \frac{2\sum e_i}{n-k} \leq n - k$.

Proving (2). Now, assume that for the selected columns it holds that $2e + s \leq n-k$. We will show that the above algorithm outputs a message $\mathbf{M} \in \mathbb{F}^{k\times k}$, such that $\Delta(\mathrm{Enc}_{\mathrm{RS}}(\mathbf{M}), \mathbf{C}^*) \leq \frac{(n-k)^2}{2}$. To prove this, observe that since $2e+s \leq n-k$ it must be the case that for every $i \in [k]$ there exists a Reed-Solomon code $\mathbf{W}^*_i \in \mathrm{RS}[k, n - s, \mathbb{F}]$ such that $\Delta(\mathbf{W}^*_i, \mathbf{W}_i) < (n - k - s)/2$. Hence, each $\mathbf{W}_i$ uniquely decodes to a message $\mathbf{M}'_i$. Furthermore, it must hold that $\mathbf{M}'_i = \mathbf{M}_i$ since their RS codewords must agree on all "correct" columns (selected by the algorithm) and the number of these columns is larger than k. To prove that $n - s - e > k$, where $n - s - e$ the number of "correct" columns, observe that $n - s - e \geq n - s - (n - s - k)/2$. Furthermore, given that $s < n - k$, it follows that $n - s - (n - s - k)/2 > (n - (n - k)) - (n - k - (n - k))/2 = k$.

De-Randomizing the Decoding Algorithm. Note that we can make the above algorithm deterministic using the same discretization method as in Forney's algorithm for concatenated codes. Note that in our case $Q = \{0, 2/(n - k), 4/(n - k), \ldots, 1\}$.

Note that the above algorithm can be naturally generalized for RS tensor-codes $\mathrm{RS}[(k_1, k_2), (n_1, n_2), \mathbb{F}]$ where $k_1 \neq k_2$ and $(n_1 \neq n_2)$.

2 Proof of Correlated Agreement of $\mathrm{RS}^{\otimes 2}$ over Curves

Theorem 2. *[Correlated Agreement of $RS^{\otimes 2}$ over Lines] Let $RS^{\otimes 2}[(k_1, n_1), (k_2, n_2), \mathbb{F}]$ be an RS tensor-code composed by the RS codes RS_1, RS_2 (presenting $(\frac{n_{RS_i}}{|\mathbb{F}|}, \frac{\delta_{RS_i}}{2})$ correlated agreement for $i = \{1, 2\}$ [2]). Additionally, let $\boldsymbol{W}_1$, $\boldsymbol{W}_2 : \mathbb{H}_1 \times \mathbb{H}_2 \to \mathbb{F}$ be some functions and $\mathcal{S}$ be a set such that:*

$$\mathcal{S} = \{a \in \mathbb{F} : \Delta(\boldsymbol{W}_1 + a\,\boldsymbol{W}_2, RS^{\otimes 2}) < \delta_{RS^{\otimes 2}}/2\}$$

If $|\mathcal{S}| > n_1 \cdot n_2 \cdot 2 \cdot \max(n_{RS_1}, n_{RS_2})$, then, $\mathcal{S} = \mathbb{F}$. Also, there exists $\boldsymbol{C}_1, \boldsymbol{C}_2 \in RS^{\otimes 2}$ and a set D where $\frac{|D|}{n_1 n_2} > 1 - \delta_{RS^{\otimes 2}}$ such that $(\boldsymbol{W}_1[d], \boldsymbol{W}_2[d]) = (\boldsymbol{C}_1[d], \boldsymbol{C}_2[d])$ for all $d \in \mathcal{D}$.

Here we adapt the above Theorem 2 to work over curves. In particular we state the following:

Theorem 3. *[Correlated Agreement of $RS^{\otimes 2}$ over Parametrized Curves] Let $RS^{\otimes 2} = RS_1 \otimes RS_2$, and assume that RS_i presents $(\frac{d \cdot n_{RS_i}}{|\mathbb{F}|}, \frac{\delta_{RS_i}}{2})$ correlated agreement over parametrized curve. Furthermore, let $\boldsymbol{W}_1, \ldots, \boldsymbol{W}_d : \mathbb{H}_1 \times \mathbb{H}_2 \to \mathbb{F}$ be some functions and $\delta < \frac{(n_1 - k_1)(n_2 - k_2)}{2} = \frac{\delta_{RS^{\otimes 2}}}{2}$ and $\mathcal{S}$ be a set such that:*

$$\mathcal{S} = \{a \in \mathbb{F} : \Delta(\sum_{i \in [d]} a^{i-1}\,\boldsymbol{W}_i, RS^{\otimes 2}) < \delta\}$$

If $|\mathcal{S}| > 2 \cdot n_1 \cdot n_2 \cdot \max(d \cdot n_{RS_1}, d \cdot n_{RS_2})$, then, $\mathcal{S} = \mathbb{F}$. In addition, there exist $\boldsymbol{C}_1, , \ldots, \boldsymbol{C}_d \in RS^{\otimes 2}$ and a set D where $\frac{|D|}{n_1 n_2} > 1 - \delta_{RS^{\otimes 2}}$ such that: (1) $\Delta(\sum_{i \in [d]} a^{i-1}\,\boldsymbol{W}_i, \sum_{i \in [d]} a^{i-1}\,\boldsymbol{C}_i) < \delta$ for all $a \in \mathbb{F}$ and $(\boldsymbol{W}_1[d], \ldots, \boldsymbol{W}_d[d]) \neq (\boldsymbol{C}_1[d], \ldots, \boldsymbol{C}_d[d])$ for all $d \in \mathcal{D}$.

To prove the above, we use an almost identical proof with that of Theorem 2. Note that Claim 1 as well as its proof remains the same with the minor difference that $|\mathcal{S}|$ is now larger than $n_1 \cdot n_2 \cdot n_{\mathrm{RS}} \cdot d$. Furthermore, we use the same simplifications, since, Corollary 1.3 of [2] asserts that for each column, there must be at most $d n_{\mathrm{RS}}$ values that do not have the same distance δ_i. Next, step 3 of our decoding algorithm by having it select a θ with at least $d \cdot n_2 \cdot n_{\mathrm{RS}}$ successful decodings. Note that such θ exists since the number of columns remains the same and we only multiplicatively increase $\mathcal{S}$ by a factor of d. Finally, steps 4 and 5 are adapted to decode d matrices $\mathbf{U}_1, \ldots, \mathbf{U}_d$ and $\mathbf{M}_1, \ldots, \mathbf{M}_d$. We finally follow the same argument as in Lemma 1 to prove theorem 3.

3 Obtaining Zero-Knowledge on FRI-like Protocols Without Masking

In Sections 3.1, 3.2 we show how to construct such an zero-knowledge RS code specifically for proofs produced by the FRI protocol, and in Section 3.3 extend

our techniques to construct a (t, l)-zero knowledge collaborative RS tensor-codes with the same properties.

3.1 Zero-Knowledge RS Codes for FRI

Here, we construct an L-zero-knowledge RS code such that the proof obtained when running the FRI protocol is inherently zero-knowledge. Recall that an FRI proof consists of: (1) the oracles $[\mathbf{c}^{(1)}], \ldots, [\mathbf{c}^{(k)}]$, where $\mathbf{c}^{(1)}$ the original zero-knowledge codeword and $\mathbf{c}^{(i)}$ the folding of $\mathbf{c}^{(i-1)}$ (e.g., $\mathbf{c}^{(i)} = \text{Fold}(\mathbf{c}^{(i-1)}, r_{i-1})$ where r_{i-1} a verifier's challenge) and (2) L query replies $\mathbf{c}^{(1)}[\mathcal{I}_1], \ldots, \mathbf{c}^{(k)}[\mathcal{I}_k]$ where $\mathcal{I} = \mathcal{I}_1 \cup \ldots \mathcal{I}_k$ an index query set following a specific query pattern which we will describe later. With that in mind, we wish to prove the following:

Theorem 4 (Honest-Verifier Zero-Knowledge RS Codes for FRI Queries)**.** *There exists a L-zero-knowledge RS code $RS_r = RS[2k, n, \mathbb{F}]$ and a PPT simulator $\mathcal{S}$ such that for any adversary $\mathcal{A}$ and message $\boldsymbol{x} \in \mathbb{F}^k$, the following distributions are statistically indistinguishable with error $\epsilon_{zk} = negl(\lambda)$ for large enough $\mathbb{F}$:*

1. *The real view $\text{View}^{\mathcal{R}}_{\mathcal{A}}(\boldsymbol{x})$ of $\mathcal{A}$ which is $(\{[\boldsymbol{c}^{(i)}]\}_{i\in[k]}, \{\boldsymbol{c}^{(i)}[\mathcal{I}_i]\}_{i\in[k]})$, where $|\mathcal{I}_1 \cup \ldots \cup \mathcal{I}_k| = L$.*
2. *The simulated view $\text{View}^{\mathcal{S}}_{\mathcal{A}}$ of $\mathcal{A}$ where all queries are answered by $\mathcal{S}$.*

Query Pattern $\mathcal{I}$. In this section we are specifically interested in the query pattern produced by the FRI protocol. In particular $\mathcal{I}_1$ (with $|\mathcal{I}_1| = 2l$) consists of l pairs of the form $(x, -x)$ (where $-x = x + |\mathbf{c}^{(1)}|/2$). In other words, $\mathcal{I}_1$ is formed in such that the verifier can locally compute l elements of the vector $\mathbf{c}^{(2)}$ at the indexes $\mathcal{I}'_1 = \{x_1, \ldots, x_l\}$ for all $(x_i, -x_i) \in \mathcal{I}_1$. Next, we define each $\mathcal{I}_i$ (with $|\mathcal{I}_i| = l$) as $\mathcal{I}_i = \{-x, \forall x \in \mathcal{I}'_i\}$ where $\mathcal{I}'_i$ the set of "folded" indexes produced by $\mathcal{I}_{i-1}$ and $\mathcal{I}'_{i-1}$. Finally, we assume that $|\mathbf{x}^{*(k)}| = 2l$ and thus $k = \log \frac{d}{l}$. Note that $L = (k+1)l$ is the total number of queries.

Our Zero-Knowledge RS Code. Based on the above pattern, we now propose our L-zero-knowledge RS code. In particular we will focus on constructing an encoding algorithm $\text{Enc}_{\text{RS},r} : \mathbb{F}^d \times \mathbb{F}^L \to \mathbb{F}^n$ that takes as input a message $\mathbf{x} \in \mathbb{F}^d$ and using a random vector $\mathbf{x}_r \in \mathbb{F}^L$ outputs a zero-knowledge codeword $\mathbf{c} \in \mathbb{F}^n$. Note that the standard approach of constructing such an algorithm is to first initialize a (randomized) polynomial $p : \mathbb{F}^{<d+L} \to \mathbb{F}$ the coefficients of which are $(\mathbf{x}, \mathbf{x}_r)$ and then output $\mathbf{c}$ by evaluating p on all points of a multiplicative coset $\mathbb{H}_1$ of $\mathbb{F}^*$. Unfortunately, and as highlighted in [8], while this approach hides L queries performed directly over $\mathbf{c}$ it is unclear wether it can also hide queries of foldings of $\mathbf{c}$—which is necessary for proving Theorem 4.

Instead, we follow a different approach. In particular, our algorithm uses $\mathbf{x}$ and $\mathbf{r}$, to form a padded (with randomness) vector $\mathbf{x}^* \in \mathbb{F}^{2d}$ and, instead of setting $\mathbf{x}^*$ to be the coefficients of p, we set p to be the interpolation of $\mathbf{x}^*$ at the points $\mathbb{H}_2$ where $\mathbb{H}_2 \cap \mathbb{H}_1 = \emptyset$ and $\mathbb{H}_2$ a multiplicative coset of $\mathbb{F}^*$. By properly organizing $\mathbf{x}^*$, the above will enable us to prove zero-knowledge under folding.

What remains is to show how we actually form $\mathbf{x}^*$. First, we segment $\mathbf{x}_r$ in l-sized segments $(\mathbf{x}_{r,1}, \ldots, \mathbf{x}_{r,k+1})$ and $\mathbf{x} = (\mathbf{x}_l, \mathbf{x}_r) \in \mathbb{F}^l \times \mathbb{F}^{d-l}$. Then, we set $\mathbf{x}^* = (\mathbf{x}^*_L, \mathbf{x}^*_R) \in \mathbb{F}^d \times \mathbb{F}^d$, where $\mathbf{x}^*_R = (\mathbf{x}, \mathbf{x}_{r,k+1}) \in \mathbb{F}^l$ and $\mathbf{x}^*_L$ is zero everywhere except from $(k+1)$ l-sized segments: $\mathbf{x}^*_{L,2^i} = \mathbf{x}_{r,i}$ for all $i \in [k]$ and $\mathbf{x}^*_{L,3} = \mathbf{x}_l$ (see Figure 1 we provide a visual representation of $\mathbf{x}^*$). Note that throughout this section, we assume that d is a power of two. In practice, if it hold that $d+(k+1)l$ is power of two, then, no padding to the next power of two is needed!

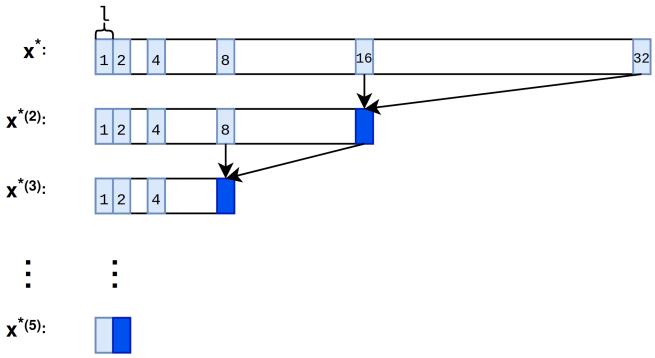


Figure 1: This Figure illustrates the arrangement of the randomly sampled segments in $\mathbf{x}^* \in \mathbb{F}^{32 \cdot l}$ (highlighted with cyan) as well as the arrangement of the random segments in the folded $\mathbf{x}^{*(i)}$. The numbers on top of the segments indicate the indexes. Note that when $\mathbf{x} = \mathbf{0}^d$, each segment with light-blue color on any $\mathbf{x}^{*(i)}_j$ equals to $\frac{1}{2}\mathbf{x}^{*(i-1)}_j$

Now, in order to prove Theorem 4, we first prove the following Lemma:

Lemma 1. *For any query set $\mathcal{I}$ (as defined above), messages $\boldsymbol{x}_1, \boldsymbol{x}_2$, query replies $\boldsymbol{q} \in \mathbb{F}^{(k+1)l}$, and padded vectors $\boldsymbol{x}^*_1$, $\boldsymbol{x}^*_2$ of $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ respectively, the following holds:*

$$\Pr[(\boldsymbol{c}^{(1)}_1[\mathcal{I}_1], \ldots, \boldsymbol{c}^{(k)}_1[\mathcal{I}_k]) = \boldsymbol{q}] = \Pr[(\boldsymbol{c}^{(1)}_2[\mathcal{I}_1], \ldots, \boldsymbol{c}^{(k)}_2[\mathcal{I}_k]) = \boldsymbol{q}]$$

With probability greater than $1-negl(\lambda)$ over the verifier's challenges $(r_1, \ldots, r_{k-1})$, where $\boldsymbol{c}^{(1)}_i = Enc_{RS}(\boldsymbol{x}^*_i)$ for $i \in \{1, 2\}$.

Proof of Lemma 1. To prove the above, we will need to show that for any $\mathbf{x}$ and $\mathbf{q} = (\mathbf{q}^{(1)}, \ldots, \mathbf{q}^{(k)}) = (\mathbf{c}^{(1)}[\mathcal{I}_1], \ldots, \mathbf{c}^{(k)}[\mathcal{I}_k])$ there exists $\mathbf{x}_r = \{\mathbf{x}_{r,i}\}_{i \in [k+1]}$ such that $\mathbf{c}^{(1)} = \mathrm{Enc}_{\mathrm{RS}}(\mathbf{x}^*)$. In other words, we need to show invertibility of the matrix $\mathbf{M} \in \mathbb{F}^{2d \times 2d}$ obtained by the linear equations: (1) $\mathbf{x}^*[j] = \sum_{i \in [2d]} c_i \omega^i_j$ for $j \in [2d] \backslash \mathcal{I}_z$ (where $\mathcal{I}_z$ the set of indexes where we place the random elements $\mathbf{x}_{r,i}$) and (2) $\mathbf{q}^{(k)}[j] = \sum_{i \in [2d]} c_i \cdot \mathbf{v}^{(k,j)}_i(r_1, \ldots, r_{k-1})$, where $\mathbf{v}^{(k,j)}(r_1, \ldots, r_{i-1})$ is the linear constraint that evaluates $\mathbf{q}^{(k)}[j]$[1].

[1]Note that $\mathbf{v}^{(1,j)}$ does not depend on the verifier's challenge

Note that the second set of constraints, essentially makes $\mathbf{M}$ dependent on the variables $R_1, \ldots, R_{k-1}$ (corresponding to the verifier's challenges) and thus it can be seen as a function $\mathbf{M}(R_1, \ldots, R_{k-1}) : \mathbb{F}^{k-1} \to \mathbb{F}^{2d \times 2d}$. In addition, $\det(\mathbf{M}(R_1, \ldots, R_{k-1}))$ is a $(k-1)$-variate polynomial $f_{\det} : \mathbb{F}^{k-1} \to \mathbb{F}$ of individual degree less than lk. Note that if $f_{\det}$ is a non-zero polynomial, then, due to the Schwartz-Zippel lemma, with probability $1 - \frac{lk^2}{|\mathbb{F}|}$ over a random $\mathbf{r} = (r_1, \ldots, r_{k-1})$, $\det(\mathbf{M}(r_1, \ldots, r_{k-1})) \neq 0$ meaning that $\mathbf{M}(r_1, \ldots, r_{k-1})$ is invertible and thus Equation of Lemma 1 holds.

What remains is to show that $f_{\det}(R_1, \ldots, R_{k-1})$ is actually non-zero. For this, it is enough to find a specific $\mathbf{r}$ such that $\det(\mathbf{M}(\mathbf{r})) \neq 0$. To this end, we will set $\mathbf{r} = 0$ and show that $\det(\mathbf{M}(0, \ldots, 0)) \neq 0$.

Proving that $\det(\mathbf{M}(0, \ldots, 0)) \neq 0$. Let $\mathbf{M}_0 = \mathbf{M}(0, \ldots, 0)$. We will show that the system $\mathbf{M}_0 \cdot \mathbf{c} = \mathbf{0}^{2d}$ has a single solution $\mathbf{c} = \mathbf{0}^{2d}$ or in other words, $\mathbf{x}^* = \mathbf{0}^{2d}$. First, observe that from the above system of equations we have that $\mathbf{x}^*$ is everywhere zero (and thus $\mathbf{x} = \mathbf{0}^d$) except from indexes where $\mathbf{x}_{i,r}$ are placed (i.e., the segments $\mathbf{x}^*_{2^i}$ for all $i \in [k+1]$). For the rest of this proof, we have to show that these segments must be equal to zero too. Since the interpolation set $\mathbb{H}_2$ is a coset of $\mathbb{F}^*$, $\mathbf{x}^* = \mathbf{x}^{*(1)}$ is foldable. We denote with $\mathbf{x}^{*(i)} = \text{Fold}(\mathbf{x}^{*(i-1)}, 0) = \frac{1}{2}\big(\mathbf{x}_L^{*(i-1)} + \mathbf{x}_R^{*(i-1)}\big)$, where $\mathbf{x}^{*(i-1)} = (\mathbf{x}_L^{*(i-1)}, \mathbf{x}_R^{*(i-1)})$. As we show next, we utilize this observation to relate $\mathbf{x}^{*(i)}$ with $\mathbf{x}^*$.
Since $(r_1, \ldots, r_{k-1}) = \mathbf{0}^{k-1}$ and $\mathbf{x} = \mathbf{0}^d$, we have that $\mathbf{x}^{*(2)} = \frac{1}{2}(\mathbf{x}^*_R + \mathbf{x}^*_L) = \frac{1}{2}(\mathbf{x}^*_R + (\mathbf{0}, \mathbf{x}_{r,k+1}), = \frac{1}{2}(\mathbf{x}^*_R[: k-l], \mathbf{x}_{r,k} + \mathbf{x}_{r,k+1}) \in \mathbb{F}^k$. In general, for any $j \in \{3, \ldots, k-1\}$ we have that:

$$\mathbf{x}^{*(j)} = \frac{1}{2^{j-1}}(\mathbf{x}_R^{*(j-1)} + \mathbf{x}_L^{*(j-1)}) =$$
$$\frac{1}{2^{j-1}}(\mathbf{x}_R^{*(j-1)} + (\mathbf{0}, \sum_{i \in \{k-(j-1), k+1\}} \mathbf{x}_{r,i})) =$$
$$\frac{1}{2^{j-1}}(\mathbf{x}^*_R[: k - 2^{j-2}l], \sum_{i \in \{k-(j-2), k+1\}} \mathbf{x}_{r,i})$$

And finally, $\mathbf{x}^{*(k)} = \frac{1}{2^{k-1}}[\mathbf{x}_{r,1}, \sum_{j \in [2,k+1]} \mathbf{x}_{r,j}] \in \mathbb{F}^{2l}$ (see Figure 1 for a visual representation of each $\mathbf{x}^{*(j)}$ when $\mathbf{x} = \mathbf{0}$).

Based on the above observation, we make the following claim:

Claim 1. *Assume that $p^{(j)}(X) = 0$ for $j \in 2, \ldots, k$ where $p^{(j)}$ the j-th folded polynomial (e.g., $p^{(j)}(X) = p_e^{(j-1)}(X) + r_{j-1}p_o^{(j-1)}(X)$ where $p_e^{(j-1)}$,$p_o^{(j-1)}$ the polynomials formed by the even and odd coefficients of $p^{(j-1)}$ respectively). Then, it must hold that $p^{(j-1)}(X) = 0$.*

Proof. Since $p^{(j)}(X) = 0$, we have that $\mathbf{x}^{*(j)} = \mathbf{0}$, and thus:

1. It must hold that $\mathbf{x}^*_R[: k - 2^{j-2}l] = \mathbf{0}$ or, in other words, $\mathbf{x}_{r,i} = \mathbf{0}^l$ for all $i \in [1, k-(j-2)]$.

2. $\sum_{i\in\{k-(j-2),k+1\}} \mathbf{x}_{r,i} = \mathbf{0}^l$.

Furthermore, observe that since $p^{(j)}(X) = p_e^{(j-1)}(X) + 0 \cdot p_o^{(j-1)}(X) = p_e^{(j-1)}(X) = 0$, then, it must hold that $p^{(j-1)}(X) = X p_o^{(j-1)}(X^2)$.

Having established that, we make the following additional observations regarding $\mathbf{x}^{*(j-1)}$ and $p^{(j-1)}(X)$:

1. Based on the observation (1), all l-sized segments of $\mathbf{x}^{*(j-1)}$ are zero, except from those in the middle and end of $\mathbf{x}^{*(j-1)}$ where they are $\mathbf{x}_{r,k-(j-2)}$ and $\sum_{i\in\{k-(j-3),k+1\}} \mathbf{x}_{r,i}$ respectively.

2. Next, the polynomial $p^{(j-1)}(X)$ of degree less than $\frac{d}{2^{j-3}}$ has $\frac{d}{2^{j-3}}+1$ roots. The first $\frac{d}{2^{j-3}} - 2l$ roots stem from the zero values of $\mathbf{x}^{*(j-1)}$ and another one derives from the fact that $p^{(j-1)}(0) = 0 \cdot p_o^{(j-1)}(0^2) = 0$. Finally, the rest of the $2l$ roots come from the fact that $p^{(j-1)}(x) = 0$, for $x \in \mathcal{I}_{j-1}$ and that $p^{(j-1)}(-x) = -p^{(j-1)}(x) = 0$. Hence we obtain $2|\mathcal{I}_{j-1}| = 2l$ additional roots.

From the last observation, we get that $p^{(j-1)}(X) = 0$ and thus $\mathbf{x}_{r,k-(j-2)}$ must be equal to $\mathbf{0}^l$. □

To complete our proof, we need to show that $p^{(k)}(X) = 0$. If this holds, then, we sequentially apply the above lemma to show that $p^{(1)}(X) = 0$. For this, note that $p^{(k)}(X)$ has degree less than $2l$ and it also has $2l$ roots. The first l roots come directly from the queries at the k level, while the other l roots come by repeatedly folding $\mathbf{c}^{(1)}$.

Proof of Theorem 4. Roughly speaking Lemma 1 states any FRI proof over our zero-knowledge code follows a uniform distribution. We use this, to construct a simulator that outputs $\mathbf{q}$ uniformly at random. First, recall that $\text{View}_{\mathcal{A}}^{\mathcal{R}}$ consists of all query replies $\mathbf{R}_i = \mathbf{c}^{(i)}[\mathcal{I}_i]$ for each round $i \in [k]$ as well as the oracles of $\mathbf{c}^{(i)}$. Hence, $\text{View}_{\mathcal{A}}^{\mathcal{R}} = (\{[\mathbf{c}^{(i)}]\}_{i\in[k-1]}, \{\mathbf{R}_i\}_{i\in[k]})$.

With that in mind we construct a simulator $\mathcal{S}$ that has access to the random tape of $\mathcal{V}$ and works as follows:

1. Initializes the verifier's queries by setting $\mathbf{R}_1^* \xleftarrow{\$} \mathbb{F}^{2l}$ and $\mathbf{R}_i^* \xleftarrow{\$} \mathbb{F}^l$ for all $i \in [1, k]$.

2. Initializes each oracle $\mathbf{c}^{*(i)}$ by setting all elements to zero except from those queried by the verifier where are set based on $\mathbf{R}_1^*, \ldots, \mathbf{R}_i^*$. Note that for $\mathbf{c}^{*(k)}$, the simulator has $2l$ elements so it can compute it honestly.

3. Outputs $(\{[\mathbf{c}^{*(i)}]\}_{i\in[k]}, \{\mathbf{R}_i^*\}_{i\in[k]})$

What remains, is to show that for any distinguisher $\mathcal{D}$, $|\Pr[\mathcal{D}(tr)|tr \leftarrow \text{View}_{\mathcal{A}}^{\mathcal{S}}] - \Pr[\mathcal{D}(tr)|tr \leftarrow \text{View}_{\mathcal{A}}^{\mathcal{R}}]| < \text{negl}(\lambda)$. For this observe that by definition the oracles are indistinguishable from the real and ideal world so we focus on the indistinguishability of $\{\mathbf{R}_i^*\}_{i\in[k]}$ and $\{\mathbf{R}_i\}_{i\in[k]}$. Note that $\{\mathbf{R}_i^*\}_{i\in[k]}$ follows a uniform

distribution over $\mathbb{F}^{(k+1)l}$. The same holds with $\{\mathbf{R}_i\}_{i\in[k]}$ as well but with probability $1-\frac{lk^2}{|\mathbb{F}|}$. Consequently, both distributions are perfectly indistinguishable with overwhelming probability over the verifier's challenges.

3.2 Zero-Knowledge RS Codes for WHIR

Here, we focus on proving zero-knowledge of a protocol the query indexes $\mathcal{I}$ of which follow a similar pattern to that the WHIR protocol [1] (excluding the sumcheck protocol). To prove this we follow similar techniques with the ones used for the FRI protocol but the assumptions regarding the query pattern significantly differ.

Notation. First, we assume that the prover folds the codewords using an m-wise folding scheme. Let k be the number of codewords $\{\mathbf{c}^{(i)}\}_{i\in[k]}$ where $\mathbf{c}^{(1)}$ is codeword of the masked message $\mathbf{x}^*$ while $\mathbf{c}^{(i)}$ is a codeword for the folded message $\mathbf{x}^{*(i)}$, for $i>1$.

Assumptions Regarding Queries. Let $\mathcal{I}=\mathcal{I}_1\cup\ldots\cup\mathcal{I}_k$ be the set of query indexes where $\mathcal{I}_i$ corresponds to the query indexes over $\mathbf{c}^{(i)}$. We assume that $|\mathcal{I}_1|=\ldots=|\mathcal{I}_{k-1}|=ml$. Furthermore, we assume that each $\mathcal{I}_i$ consists of l tuples of the form $(i_1,\ldots,i_m)$ such that $\mathbf{c}^{(i+1)}[\tilde{i}]=\text{Fold}_m((\mathbf{c}^{(i)}[i_1],\ldots,\mathbf{c}^{(i)}[i_m]),r_i)$. In addition, we assume that elements of $\mathbf{c}^{(i)}$ with indexes $\mathcal{I}_i$ can not be folded after the $(i+1)$ level. Finally, we assume that $|\mathbf{x}^{*(k)}|=ml$ and thus $k=1+\log_m\frac{2k}{ml}$.

Note that due to the above assumptions, $|\mathcal{I}_k|$ is not ml but rather $ml-l=(m-1)l$ (due to the folding of the level $k-1$).

Our Zero-Knowledge RS code. The encoding algorithm of our zero-knowledge RS code works in the same way as the one of Section 3.1 but it organizes the padded vector $\mathbf{x}^*\in\mathbb{F}^{2k}$ in a slightly different way. Let $\mathbf{x}_r\in\mathbb{F}^L$ be the random vector where $L=kml-l$ the total number of queries.

As a first step, the algorithm parses $\mathbf{x}^*$ into l-sized segments $(\mathbf{x}^*_1,\ldots,\mathbf{x}^*_{\frac{2k}{l}})$ and similarly to the FRI-based queries, randomly sample the segments with indexes $\frac{2k}{l}$ and $\{m^i,2m^i,\ldots,(m-1)m^i\}_{i\in[1,k-1]}$. This, however, leads to only $(k-1)(m-1)l+l=kml-(m-1)l-kl+l$ random elements.

In addition, the prover masks the first $m-1$ segments, thus further increasing the randomness to $kml-(k-1)l$ elements. Finally, the prover randomly samples all elements with indexes $\{2m^i+1\}_{i\in[k-2]}$—leading to $kml-l$ random elements. Figure 2 presents the arrangement of the random elements within $\mathbf{x}^*$.
Having established that, we can prove the following:

Theorem 5. *For any query set $\mathcal{I}$ (as defined above), original messages $\boldsymbol{x}_1,\boldsymbol{x}_2$ and $\boldsymbol{q}\in\mathbb{F}^{(k+1)l}$, and mask vectors $\boldsymbol{x}_1^*$, $\boldsymbol{x}_2^*$ of $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ respectively (sampled using the algorithm described above), the following holds:*

$$\Pr[(\boldsymbol{c}_1^{(1)}[\mathcal{I}_1],\ldots,\boldsymbol{c}_1^{(k)}[\mathcal{I}_k])=\boldsymbol{q}]=\Pr[(\boldsymbol{c}_2^{(1)}[\mathcal{I}_1],\ldots,\boldsymbol{c}_2^{(k)}[\mathcal{I}_k])=\boldsymbol{q}]$$

With probability greater than $1-negl(\lambda)$ over the verifier's challenges $(\boldsymbol{r}_1,\ldots,\boldsymbol{r}_{k-1})$, where $\boldsymbol{c}_i^{(1)}=Enc_{RS}(\boldsymbol{x}_i^*)$ and $\boldsymbol{c}_i^{(j)}=Fold_m(\boldsymbol{c}_i^{(j-1)},\boldsymbol{r}_{j-1})$ for $i\in\{1,2\}$ and $\boldsymbol{r}_{j-1}\in\mathbb{F}^m$.

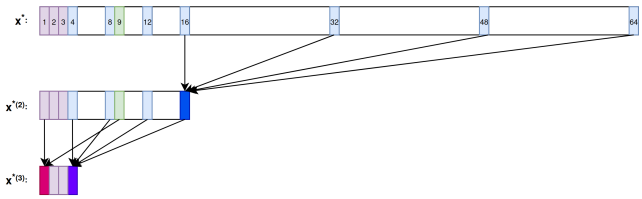


Figure 2: Arrangement of random segments when $|\mathbf{x}^*| = 64l$ and $m = 4$. With cyan color we have the segments placed by the first step of the algorithm, purple the segments inserted by the second step and green the segments inserted by the third step.

Proof of Theorem 5. Let $\mathbf{M}(\vec{R})$ be the $2d \times 2d$ system of linear equations as defined in Section 3.1, where $\vec{R}$ contains mk random variables. We will show that $\mathbf{M}(\mathbf{r}_1, \ldots, \mathbf{r}_k)$ is invertible with probability $\frac{(mkl)^2}{|\mathbb{F}|}$ by showing that $\mathbf{M}_0 = \mathbf{M}(\mathbf{b}_1, \ldots, \mathbf{b}_{k-1})$ is invertible. Here $\mathbf{b}_i$ is defined as $\mathbf{b}_i = (\beta(1, \mathbf{0}), \ldots, \beta(m, \mathbf{0}))$, where $\mathbf{0}$ contains $\log m$ zeroes. In this way, the function Fold_m can be written as $\text{Fold}_m = \text{Fold}_2(\text{Fold}_2(..., 0), 0)$ [1]. As we will see later this will be crucial for proving invertibility of $\mathbf{M}_0$.

With that in mind, we will prove that the system $\mathbf{M}_0 \cdot \mathbf{c} = \mathbf{0}$ has a unique solution, $\mathbf{c} = 0$. First, note that the above system of equations implies that $\mathbf{x}^*$ is zero everywhere except from the random segments. Here we will show that these segments must also be zero.

Let $d_i = |\mathbf{x}^{*(i)}|$. Observe that $\mathbf{x}^{*(2)}$ can be written as $\mathbf{x}^{*(2)} = (\frac{1}{m}\mathbf{x}^*[d_2 - l], \text{Fold}_m((\mathbf{x}_{\frac{d_2}{l}}, \ldots, \mathbf{x}_{\frac{md_2}{l}}), 0))$. Furthermore, for every $j > 2$ we can write $\mathbf{x}^{*(j)}$ as $(\mathbf{x}_1^{*(j)}, \frac{1}{m^{j-1}}\mathbf{x}^*[l : d_j - l], \mathbf{x}^{*(j)}_{\frac{d_j}{l}})$ where:

- $\mathbf{x}_1^{*(j)} = \text{Fold}_m(\mathbf{x}_1^{*(j-1)}, \mathbf{0}^{l(m-2)/2}, \mathbf{x}^{*(j-1)}_{\frac{m}{2}\frac{d_j}{l}}, \mathbf{0}^{l(m-2)/2}) = \frac{2}{m}\text{Fold}_2(\mathbf{x}_1^{*(j-1)}, \mathbf{x}^{*(j-1)}_{\frac{md_j}{2l}})$
- $\mathbf{x}^{*(j)}_{\frac{d_j}{l}} = \text{Fold}_m(\mathbf{x}^{*(j-1)}_{\frac{d_j}{l}}, \mathbf{x}^{*(j-1)}_{2\frac{d_j}{l}}, \ldots, \mathbf{x}^{*(j-1)}_{m\frac{d_j}{l}})$

With that in mind, we introduce the following lemma:

Lemma 2. *Assume that $p^{(j+1)} \equiv 0$ for $j \geq 2$. Then, $p^{(j)} \equiv 0$.*

Proof. First, note that since $p^{(j+1)} \equiv 0$, it holds that $\mathbf{x}^{*(j+1)} = \mathbf{0}$. Next, by the special structure of $\mathbf{b}_{j-1}$ we can define the intermediate foldings $\mathbf{x}^{*(j,1)}, \ldots, \mathbf{x}^{*(j,\log m)}$ as $\mathbf{x}^{*(j,i)} = \text{Fold}_2(\mathbf{x}^{*(j,i+1)}, 0)$ where $\mathbf{x}^{*(j,1)} = \mathbf{x}^{*(j)}$ and $\mathbf{x}^{*(j,\log m)} = \mathbf{x}^{*(j+1)}$. Let $p^{(j,1)}, \ldots, p^{(j,\log m)}$ be the polynomials that interpolate these foldings with $p^{(j,\log m)} \equiv 0$.

We will first show that $\mathbf{x}^{*(j,2)} = \mathbf{0}$. First, by the above observation, $\mathbf{x}^{*(j,\log m)} = \mathbf{x}^{*(j+1)}$ has the following form:

$$(\frac{2}{m}\mathbf{x}_1^{*(j,2)}, \mathbf{x}^*[l, d_{j+1} - l], \text{Fold}_{m/2}(\mathbf{x}^{*(j,2)}_{d_{j+1}}, \mathbf{x}^{*(j,2)}_{2d_{j+1}}, \ldots, \mathbf{x}^{*(j,2)}_{\frac{m}{2}d_{j+1}}))$$

Note that since $\mathbf{x}^{*(j,\log m)} = \mathbf{0}$, and thus $\mathbf{x}^*[l, d_{j+1} - l] = 0$, it means that all random segments segments are guaranteed to be zero except from those used to produce $\mathbf{x}^{*(j,2)}_{d_{j+1}}, \mathbf{x}^{*(j,2)}_{2d_{j+1}}, \ldots, \mathbf{x}^{*(j,2)}_{\frac{m}{2}d_{j+1}}$ and $\mathbf{x}^{(j,2)}_1$. Consequently, since $\mathbf{x}^{(j,2)}_1 = \mathbf{0}$ it must hold that the only possible non-zero elements are those in $\mathbf{x}^{*(j,2)}_{d_{j+1}}, \mathbf{x}^{*(j,2)}_{2d_{j+1}}, \ldots, \mathbf{x}^{*(j,2)}_{\frac{m}{2}d_{j+1}}$. In other words $\mathbf{x}^{*(j,2)}$ is zero everywhere except from possibly $\frac{m}{2}l$ elements which means that the $\frac{m}{2}d_{j+1}$-degree polynomial $p^{(j,2)}$ at least $\frac{m}{2}d_{j+1} - \frac{m}{2}l$ roots. In order to prove that $p^{(j,2)} \equiv 0$ and thus $\mathbf{x}^{(j,2)} = 0$, we need to find another $\frac{m}{2}l$ roots. We can find these roots by observing that $\mathbf{c}^{(j)}[\mathcal{I}_j] = \mathbf{0}^{ml}$ and thus $\text{Fold}(\mathbf{c}^{(j)}[\mathcal{I}_j], 0) = \mathbf{0}^{\frac{ml}{2}}$.

Now, given that $p^{(j,2)} \equiv 0$, we have that $p^{(j)}(X)$ is of the form $p^{(j)}(X) = Xp'(X^2)$. To complete the proof, we need to show that $p^{(j)}$ has at least d_j roots. Using the fact that $\mathbf{x}^{*(j,2)} = \mathbf{0}$, it follows that $\mathbf{x}^{*(j)}$ has $d_j - l(m+2)$ zeroes and thus $p^{(j)}$ has $d_j - l(m+2)$ roots. In addition, $p^{(j)}$ has lm additional roots since $p^{(j)}(x) = 0$ for all $x \in \mathcal{I}_j$. Finally, $p^{(j)}$ has $2l$ roots from the fact that $p^{(j)}(x') = 0 = p^{(j)}(-x')$, where x' is a folded index of $\mathcal{I}_{j-1}$. □

To complete the proof, we need to show that (1) $p^{*(k)} \equiv 0$ (or in other words $\mathbf{x}^{*(k)} = \mathbf{0}$) and (2) if $p^{*(2)} \equiv 0$, then, we must have that $p^{*(1)} \equiv 0$. (1) comes directly from the fact that $p^{*(k)} \equiv 0$ has ml roots and its degree is less than ml. As for (2), observe that $\mathbf{x}^{*(2)} = (\frac{1}{m}\mathbf{x}^*[\frac{2d}{m} - 1], \text{Fold}_m((\mathbf{x}_{\frac{2d}{ml}}, \ldots, \mathbf{x}_{\frac{2d}{l}}), 0)) = \mathbf{0}$ and thus $\mathbf{x}^*[\frac{2d}{m} - 1] = \mathbf{0}$. Consequently, $\mathbf{x}^*$ will have $2d - ml$ zeroes. Given that $p^{(1)}(x) = 0$, for all $x \in \mathcal{I}_1$, we conclude that $p^{(1)}(X)$ has $2d$ roots and since its degree is less than $2d$, it must hold that $p^{(1)} \equiv 0$.

Handling a Different Number of Queries. Currently, we assumed that on each folding step, the verifier performs the same number of queries. However, this is not always true for WHIR-based protocols. One simple way to overcome this limitation would be to set $l = \max(l_1, \ldots, l_k)$ where l_i the number of queries asked by the verifier during the i-th round. With overwhelming probability, this will result in a underdetermined linear system and thus equation of Theorem 5 would hold.

Handling a Different Number of Queries. In the above proof, we also assumed the the folding degree on each step is constant and equal to m. However, we would like to support different folding degrees for each step. To support this, we have to slightly modify the way we assign the random segments to $\mathbf{x}^*$. Let $m_1, \ldots, m_{k-1}$ be powers of two integers where m_i corresponds to the folding degree of the i-th folding. We assign with randomness the following index segments: (1) the first $m_{k-1} - 1$ segment indexes of $\mathbf{x}^*$ (2) all segments of $\mathbf{x}^*$ with indexes $\{M^{(i)}, m_i M^{(i)}, \ldots, (m_i - 1)M^{(i)}\}$ where $M^{(i)} = \prod_{j \in \{i, k-1\}} m_i$ and finally (3) $\{2m_i M^{(i)} + 1\}_{i \in [k-2]}$.

Essentially, we organize $\mathbf{x}^*$ in such a way so that $\mathbf{x}^{*(2)}, \ldots, \mathbf{x}^{*(k)}$ will have the same form as in the above proof which will enable us to prove lemma 2.

3.3 Adapting our Techniques in the Collaborative Setting

Here, we show how to adapt the techniques of the previous section to construct a (t, L)-zero-knowledge collaborative $\mathrm{RS}^{\otimes 2}$ code such that, when running a collaborative FRI-like protocol over it, we obtain t-zero-knowledge for free, without the need to perform any masking. In particular, we prove the following:

Theorem 6. *There exists a $(t, (k+1)l)$-zero-knowledge collaborative code $RS^{\otimes 2}[(d, d+t, N), (m, 2m, M), \mathbb{F}]$ and a PPT simulator $\mathcal{S}$ such that for any adversary $\mathcal{A}$ that controls at most t provers $\{\mathcal{P}_{i_1}, \ldots, \mathcal{P}_{i_t}\}$, where $(i_1, \ldots, i_t) \in \mathcal{C}$ and the verifer, and any "global" input $\boldsymbol{X} \in \mathbb{F}^{d \times m}$, the following distributions are statistically indistinguishable with error $\epsilon_{zk} = lk^2/|\mathbb{F}|$:*

1. *The real view $\mathrm{View}_{\mathcal{A}}^{\mathcal{R}}$ of $\mathcal{A}$ where all queries are answered by the oracles $(\boldsymbol{C}^{(1)}, \boldsymbol{C}^{(2)}, \ldots, \boldsymbol{C}^{(k)})$.*
2. *The simulated view $\mathrm{View}_{\mathcal{A}}^{\mathcal{S}}$ of $\mathcal{A}$ where all queries are answered by $\mathcal{S}$.*

We begin by describing our $\Pi^{\mathrm{Enc}_{\mathrm{RS}^{\otimes 2}}}$ functionality and explain how to instantiate it in practice. Then, we use it to prove the above theorem. Finally, we also propose an optimization which concretely reduces the encoding time of each prover by half.

The Functionality $\Pi^{\mathbf{Enc}_{\mathbf{RS}^{\otimes 2}}}$. Our functionality receives as input the individual input $\mathbf{X}_i \in \mathbb{F}^{d \times \frac{m}{N}}$ of each prover $\mathcal{P}_i$ and concatenates all inputs forming $\mathbf{X} \in \mathbb{F}^{d \times m}$. Next, it embeds $d + t$ random rows at the end of $\mathbf{X}$ and then pads each row forming the matrix $\mathbf{X}^* \in \mathbb{F}^{(d+t) \times 2m}$. To pad each row, the functionality uses the same padding technique as described in Section 3.1. Finally, the functionality constructs the bivariate polynomial $p : \mathbb{F}^2 \to \mathbb{F}$ by interpolating $\mathbf{X}^*$ and invokes $\mathrm{Enc}_{\mathrm{RS}^{\otimes 2}}(p)$ to obtain $\mathbf{C} \in \mathbb{F}^{N \times M}$ and sends $\mathbf{C}_i$ to $\mathcal{P}_i$. In order to instantiate this functionality in practice, we follow a protocol identical to that of [12], but the provers, instead of padding each row of $\mathbf{X}$ by concatenating the random columns, they follow the padding strategy of Section 3.1.

Having established that, we can now prove the following Lemma which is crucial for proving Theorem 6:

Lemma 3. *Let $\mathcal{C} \in [N]^t$ be a set where $t \leq d' - d$. Furthermore, let $\boldsymbol{C}_1^{(1)}, \boldsymbol{C}_2^{(1)}$ the $RS^{\otimes 2}[(d, d', N), (m, 2m, M), \mathbb{F}]$ codewords that encode $\boldsymbol{X}_1 = (\boldsymbol{X}_{1,1}, \ldots, \boldsymbol{X}_{1,N})$ and $\boldsymbol{X}_2 = (\boldsymbol{X}_{2,1}, \ldots, \boldsymbol{X}_{2,N})$ respectively such that $\boldsymbol{X}_{1,j} = \boldsymbol{X}_{2,j}$ for all $j \in \mathcal{C}$. For any $\boldsymbol{X}_1, \boldsymbol{X}_2$, point query set $\mathcal{I} = \mathcal{I}_1 \cup \ldots \cup \mathcal{I}_k$ where $\mathcal{I}$ is defined as in Section 3.1, point query replies $\boldsymbol{Q} \in \mathbb{F}^{N \times (k+1)l}$, and t RS codewords $\boldsymbol{Q}' \in \mathbb{F}^{t \times M}$ that agree with $\boldsymbol{Q}$ in the respective indexes the following holds:*

$$\Pr[(\{\boldsymbol{C}_{1,i}^{(1)}\}_{i \in \mathcal{C}}, \{\boldsymbol{C}_1^{(i)}[\mathcal{I}_i]\}_{i \in [k]}) = (\boldsymbol{Q}', \boldsymbol{Q})] =$$
$$\Pr[(\{\boldsymbol{C}_{2,i}^{(1)}\}_{i \in \mathcal{C}}, \{\boldsymbol{C}_2^{(i)}[\mathcal{I}_i]\}_{i \in [k]}) = (\boldsymbol{Q}', \boldsymbol{Q})]$$

with probability $1 - negl(\lambda)$.

Proof. For simplicity, we prove a stronger version of Lemma 3 where the verifier does column queries and not point queries. Note that the latter case implies the former one. Having established that, our proof is similar to that of Lemma 1. In particular, given the global message $\mathbf{X} \in \mathbb{F}^{d \times m}$, $\mathbf{Q}$ and $\mathbf{Q}'$, we show that there exists a unique padded message $\mathbf{X}^* \in \mathbb{F}^{d' \times 2m}$ such that $\mathbf{C} = \mathrm{Enc}_{\mathrm{RS}^{\otimes 2}}(p)$— where $p : \mathbb{F}^2 \to \mathbb{F}$ the polynomial that interpolates $\mathbf{X}^*$ at $\mathbb{H}_1 \times \mathbb{H}_2$. Note that $\mathbf{C}$ consists of all evaluations of p at $\mathbb{H}'_1 \times \mathbb{H}'_2$ where $(\mathbb{H}'_1 \times \mathbb{H}'_2) \cap (\mathbb{H}_1 \times \mathbb{H}_2) = \emptyset$. In order to compute $\mathbf{X}^*$ it is enough to find p.

For this, we decode each row of $\mathbf{Q}'$ to obtain $p(\omega'_{1,j}, X)$, for all $j \in \mathcal{C}$. Next, we parse $\mathbf{Q}$ to submatrixes $(\mathbf{Q}_1, \ldots, \mathbf{Q}_k)$ where $\mathbf{Q}_1 \in \mathbb{F}^{N \times 2l}$ and $\mathbf{Q}_i \in \mathbb{F}^{N \times l}$ and then decode each column of $\mathbf{Q}$ using the decoding algorithm of $\mathrm{RS}[d', N, \mathbb{F}]$. Let $q_i(X)$ be the polynomial that interpolates each column of $\mathbf{Q}$. We use those polynomials to find $\mathbf{Q}^* \in \mathbb{F}^{d' \times (k+1)l}$, the matrix such that each column i consists of the evaluations of q_i at $\mathbb{H}_1$.

Next, for each i-th row of $\mathbf{X}$ and $\mathbf{Q}^*$ where $i \in [d]$, we solve the system of linear equations derived by the polynomial evaluation constraints of the polynomial $p(\omega_{1,i}, X)$ and $\mathcal{I}$ which is identical to that of Lemma 1. If the system is invertible, which happens with probability at least $1 - \mathrm{negl}(\lambda)$, we obtain the polynomials $p(\omega_{1,j}, X)$ for all $j \in [d]$, where $\omega_{1,j} \in \mathbb{H}_1$. At this point, we hold the $2m$-degree polynomials $\{p(\omega'_{1,j}, X)\}_{j \in \mathcal{C}}$ and $\{p(\omega_{1,j}, X)\}_{j \in [d]}$. Since $|\mathcal{C}| + d = d'$ and all $\omega_{1,j}, \omega'_{1,j}$ are distinct, we can directly find the polynomial p. □

Proof of Theorem 6. First, observe that the view of $\mathcal{A}$ when running the protocol in the real world is $\mathrm{View}^{\mathcal{R}}_{\mathcal{A}} = (\{[\mathbf{C}^{(i)}]\}_{i \in [k]}, \{\mathbf{C}^{(1)}_i\}_{i \in \mathcal{C}}, \{\mathbf{C}^{(i)}[\mathcal{I}_i]\}_{i \in [k]})$, where $\mathcal{I}$ the random column indexes queried by $\mathcal{A}$. Observe that by definition the oracles are indistinguishable from the real world and the simulated one and also by Lemma 3, $(\{\mathbf{C}^{(1)}_i\}_{i \in \mathcal{C}}, \{\mathbf{C}^{(i)}[\mathcal{I}_i]\}_{i \in [k]})$ is uniformly distributed with a probability at least $1 - \mathrm{negl}(\lambda)$. Hence, in order to prove the above theorem we need to construct a simulator that produces a view $(\{\mathbf{C}^{(1)}_i\}_{i \in \mathcal{C}}, \{\mathbf{C}^{(i)}[\mathcal{I}_i]\}_{i \in [k]})$ following a uniform distribution. Our simulator works as follows:

1. The simulator samples t $\mathrm{RS}[2m, M, \mathbb{F}]$ codewords $\{\mathbf{C}^{(1)}_i\}_{i \in \mathcal{C}}$ uniformly at random with the only constraint that their decodings agree with what $\mathcal{P}_i$ would have computed locally for all $i \in \mathcal{C}$. This is possible since $\mathcal{S}$ has access to the random tapes of all corrupted provers and their inputs.

2. It then samples $\mathrm{RS}[d' = d + t, N, \mathbb{F}]$ codewords at random such that they satisfy the verification steps of FRI and their indexes at $\mathcal{C}$ are consistent with $\{\mathbf{C}^{(1)}_i\}_{i \in \mathcal{C}}$.

3. Instantiates the oracles $[\mathbf{C}^{(i)}]$ by setting all columns to zero except from those in $\mathcal{I}$.

4. Outputs $(\{[\mathbf{C}^{(i)}]\}_{i \in [k]}, \{\mathbf{C}^{(1)}_i\}_{i \in \mathcal{C}}, \{\mathbf{C}^{(i)}[\mathcal{I}_i]\}_{i \in [k]}$

It is not hard to see that the simulator produces a view where $(\{\mathbf{C}_i^{(1)}\}_{i\in\mathcal{C}}, \{\mathbf{C}^{(i)}[\mathcal{I}_i]\}_{i\in[k]})$ follow a uniform distribution.

4 Proof of Theorem 7 (Our coPCS)

In this section we provide a detailed proof of Theorem 7. We also provide the description of our protocol in order to assist in the evaluation of the proofs.

Theorem 7. *Our **Protocol 1** combined with the proof composition step is a coIOPP for $\mathcal{R}_{eval}$ with $\epsilon_{rbr} = \kappa_{rbr} = \max(\{\frac{2d_i}{|\mathbb{F}|}\}_{i\in[k]}, \{\frac{l_i}{|\mathbb{F}|}\}_{i\in[k]}, \{(1-\delta_{RS_i^{\otimes 2}})^{l_i}\}_{i\in[k]}, \epsilon_{prox})$ round-by-round (knowledge) soundness error and t-zero-knowledge with $\epsilon_{zk} \in \mathcal{O}(\frac{\max(\{d_i\}_{i\in[k]})k^2}{|\mathbb{F}|})$ statistical error. When making the above protocol non-interactive, the final coPCS has a per-prover complexity of $\mathcal{O}(\frac{M}{N}\log M)$, $\mathcal{O}(l\log M\log\log M + \log^2 M)$ proof size and verification time, $\mathcal{O}(k)$ interaction rounds between provers, and $\mathcal{O}(\sqrt{M})$ total communication cost (where $l = \max(\{l_i\})$ and $M = |\boldsymbol{X}|$).*

4.1 Performance Analysis

Let $M = d_0 \cdot \ldots \cdot d_k$ (where $d_k \in \mathcal{O}(N)$) be the number of the polynomial coefficients and $d_2 = \ldots = d_{k-1} = \sqrt[k-2]{M/(d_0 d_1 d_k)}$. From the instantiation of $\Pi^{\mathrm{Enc}_{\mathrm{RS}^{\otimes 2}}}$ as described in [12] we have a per-prover complexity of $\mathcal{O}(\frac{M}{N}\log\frac{M}{N})$ for the commitment phase. Regarding the evaluation phase, the sumcheck protocol (steps 2, 4.a) has a per-prover complexity of $\mathcal{O}(\frac{M}{N}\log\frac{M}{N})$ field operations (see Section 5.1), and likewise step 4.b while step 4.e requires only $\mathcal{O}(\frac{M}{N})$ field operations. So, the total per-prover complexity is $\mathcal{O}(\frac{M}{N}\log\frac{M}{N})$ field operations. As for the proof size (and likewise verification time), when we compile our coIOPP into a coPCS following using proof composition we obtain a proof size of $\mathcal{O}(l\log N + \pi_{\mathrm{whir}} + \log^2 M)$, where the $l\log N$ factor derives by querying the oracles of $\mathbf{C}, \mathbf{C}^{(1)}, \ldots, \mathbf{C}^{(k)}$, while the second part derives from the proof composition step when using the specialized sumcheck protocol of [10] to prove FFTs. Finally, the protocol requires $\mathcal{O}(k)$ communication rounds among the provers and its total communication cost is $\mathcal{O}(N\sqrt[k-2]{M/(d_0 d_1 d_k)})$ dominated by the sumcheck protocol.

4.2 Completeness Proof

Completeness proof follows directly from the completeness of the WHIR protocol as well as the completeness of the proof composition protocol of [4].

4.3 (Knowledge) Soundness Proof

We first begin by formally the relation $\mathrm{Bind}_{p,\delta_1,\delta_2}$ which will be used to show that a message $\mathbf{C}'^{(i)}$, for $i > 1$ is "well-formed".

Definition 3. *Let δ_1 and δ_2 be the proximity parameters of the tesnor-code $RS^{\otimes 2}[(d, m_1 m_2), (N, M_1 m_2), \mathbb{F}]$ and an RS code $RS[m_2, M_2, \mathbb{F}]$ respectively. We say that a function $\boldsymbol{C}'$ belongs to $Bind_{p,\delta_1,\delta_2}$ if there exists a polynomial $p : \mathbb{F}^2 \to \mathbb{F}$, a function $\boldsymbol{C}$ and a set of segment indexes $\mathcal{A}$ of size larger than $(1 - \delta_1)NM_1$ such that: (1) $\Delta(Enc_{RS}(\boldsymbol{C}[i_1, i_2]), \boldsymbol{C}'[i_1, i_2]) < \delta_2$ and (2) $\boldsymbol{C}[i_1, i_2] = Enc_{RS^{\otimes 2}}(p)[i_1, i_2]$ for all $(i_1, i_2) \in \mathcal{A}$. More formally, we define the relation $Bind_{p,\delta_1,\delta_2}$ as follows:*

$$Bind_{p,\delta_1,\delta_2} = \{[\boldsymbol{C}']; (\boldsymbol{C}', p : \mathbb{F}^2 \to \mathbb{F}) : \exists \boldsymbol{C}, \text{ and indexes } \mathcal{A} \text{ s.t. } |\mathcal{A}| > (1-\delta_1)NM_1 \wedge \Delta(Enc_{RS}(\boldsymbol{C}[i_1, i_2]), \boldsymbol{C}'[i_1, i_2]) < \delta_2 \ \wedge \ \boldsymbol{C}[i_1, i_2] = Enc_{RS^{\otimes 2}}(p)[i_1, i_2] \ \forall (i_1, i_2) \in \mathcal{A}\}$$

Lemma 4. *If $\boldsymbol{C}' \in Bind_{p, \frac{\delta_{RS^{\otimes 2}}}{2}, \frac{\delta_{RS}}{2}}$ then the polynomial $p : \mathbb{F}^2 \to \mathbb{F}$ is unique.*

Proof. Assume that there exist two p_1, p_2 such that $\mathbf{C}' \in \text{Bind}_{p_1, \frac{\delta_{\text{RS}^{\otimes 2}}}{2}, \frac{\delta_{\text{RS}}}{2}}$ and $\mathbf{C}' \in \text{Bind}_{p_2, \frac{\delta_{\text{RS}^{\otimes 2}}}{2}, \frac{\delta_{\text{RS}}}{2}}$. Let $\mathbf{C}_1$ and $\mathbf{C}_2$ be the underlying messages that satisfy the constraints of Bind and $\mathcal{A}_1, \mathcal{A}_2$ their corresponding segment index sets.

Since $\Delta(\text{Enc}_{\text{RS}^{\otimes 2}}(p_1), \text{Enc}_{\text{RS}^{\otimes 2}}(p_2)) > \delta_{\text{RS}_i^{\otimes 2}}$, $\mathcal{A}_1, \mathcal{A}_2$ can only have at most $(1-\delta_{\text{RS}_i^{\otimes 2}})NM_1$ segment indexes $\mathcal{C}$ where $\text{Enc}_{\text{RS}^{\otimes 2}}(p_1)[i_1, i_2] = \text{Enc}_{\text{RS}^{\otimes 2}}(p_2)[i_1, i_2]$ for all $(i_1, i_2) \in \mathcal{C}$. Let $\mathcal{A}_1' = \mathcal{A}_1 \backslash \mathcal{C}$ and likewise we define $\mathcal{A}_2'$. Since the encodings of the segments with those indexes have distance less than the unique decoding radius (i.e., $\delta_{\text{RS}}/2$) from the corresponding segment in $\mathbf{C}'$, then, it must hold that $\mathcal{A}_1' \cap \mathcal{A}_2' = \emptyset$. Furthermore, it must hold that $|\mathcal{A}_1'| > \frac{\delta_{\text{RS}^{\otimes 2}}}{2} NM_1$. That is because:

$$|\mathcal{A}_1| = |\mathcal{A}_1'| + |\mathcal{A}_c| > (1 - \frac{\delta_{\text{RS}^{\otimes 2}}}{2})NM_1 \equiv$$
$$|\mathcal{A}_1'| > (1 - \frac{\delta_{\text{RS}^{\otimes 2}}}{2})NM_1 - |\mathcal{A}_c| \geq$$
$$(1 - \frac{\delta_{\text{RS}^{\otimes 2}}}{2})NM_1 - |\mathcal{C}| \geq \frac{\delta_{\text{RS}^{\otimes 2}}}{2} NM_1$$

Where $\mathcal{A}_c = \mathcal{A}_1 \cap \mathcal{A}_2$.

Based on the above, we can conclude that $\Delta(\mathbf{C}'[i_1, i_2], \text{RS}) < \delta_{\text{RS}}/2$ for all segments, since $|\mathcal{A}_1| + |\mathcal{A}_2'| \geq (1 - \delta_{\text{RS}^{\otimes 2}}/2)NM_1 + (\delta_{\text{RS}^{\otimes 2}}/2)NM_1 \geq NM_1$ and $\mathcal{A}_1 \cap \mathcal{A}_2' = \emptyset$. This means that by decoding all segments of $\mathbf{C}'$ we end up with a unique $\mathbf{C}$ for which it should simultaneously hold that $\Delta(\mathbf{C}, \text{Enc}_{\text{RS}^{\otimes 2}}(p_j)) < \delta_{\text{RS}^{\otimes 2}}/2$ for $j = \{1, 2\}$, which is a contradiction. □

Next, we propose the following lemma which is crucial for proving soundness of our coIOP.

Lemma 5. *Assume that at the end of the i-th round of **Protocol 1**, where $i > 1$, the verifier holds $[\boldsymbol{C}'^{(i+1)}]$ such that $\boldsymbol{C}'^{(i+1)} \in Bind_{p^{(i+1)}, \frac{\delta_{RS_{i+1}^{\otimes 2}}}{2}, \frac{\delta_{RS_{i+1}}}{2}}$ as well as $K^{(i+1)} = \sum \hat{p}^{(i+1)}(X) \hat{p}_{\boldsymbol{b}}^{(i+1)}(X)$ and all verification steps have passed.*

Then, it must hold that $\boldsymbol{C}'^{(i)} \in Bind_{p^{(i)}, \frac{\delta_{RS_i^{\otimes 2}}}{2}, \frac{\delta_{RS_i}}{2}}$ *with probability* $1-\epsilon_{bind}$*, where* $\epsilon_{bind} = (1-\delta_{RS_i^{\otimes 2}}/2)^{l'_i} + (1-\delta_{RS_i})^{l_i - l'_i} + \frac{l_i}{|\mathbb{F}|} + \epsilon_{prox}$*, with:*

- ϵ_{prox} *the proximity gap error of* $RS_i^{\otimes 2}$*.*
- $l_i = \max(\frac{\lambda}{-\log(1-\delta_{RS_i}\delta_{RS_i^{\otimes 2}}/4)}, \frac{\lambda}{-\log(1-\delta_{RS_i}/2)} + \frac{\lambda}{-\log(1-\delta_{RS_i^{\otimes 2}}/2)})$.
- $l'_i = -\lambda/\log(1-\delta_{RS_i^{\otimes 2}}/2)$.

Alternatively, $\boldsymbol{C}^{(i)} \notin$ *Bind with probability at most* ϵ_{bind}*.*

Proof. We want to prove that, with overwhelming, there exists a polynomial $p^{(i)}$ of individual degrees $d^{(i)}$ and $m_1^{(i)} m_2^{(i)}$, a matrix $\mathbf{C}^{(i)} \in \mathbb{F}^{N \times M_1^{(i)} m_2^{(i)}}$ (formed by decoding all segments of $\mathbf{C}'^{i}$ to the closest message) and a set of segment indexes $\mathcal{A}$ of size $|\mathcal{A}| > (1 - \frac{\delta_{\mathrm{RS}_i^{\otimes 2}}}{2}) N \cdot M_1$ such that (1) $\mathrm{Enc}_{\mathrm{RS}_i^{\otimes 2}}(p^{(i)})[i_1, i_2] = \mathbf{C}^{(i)}[i_1, i_2]$ and $\Delta(\mathrm{Enc}_{\mathrm{RS}_i^{C}}(\mathbf{C}^{(i)}[i_1, i_2]), \mathbf{C}'^{(i)}[i_1, i_2]) < \frac{\delta_{\mathrm{RS}_i}}{2}$ for all $(i_1, i_2) \in \mathcal{A}$. First, recall that since the verifier passes the delegation protocol and from our initial assumption the following must hold:

1. $\mathbf{y}[z] = \mathrm{Fold}_{d_i}(\mathbf{C}^{(i)}[j_{i,z}^{(1)}, j_{i,z}^{(2)}], r_i)$ for all $z \in [l_i]$.
2. $\mathbf{C}'^{(i)}[j_{i,z}^{(1)}, j_{i,z}^{(2)} M_2 + j_{i,z}^{(3)}] = \mathrm{Enc}_{\mathrm{RS}_i^{C}}(\mathbf{C}^{(i)}[j_{i,z}^{(1)}, j_{i,z}^{(2)}])[j_{i,z}^{(3)}]$ for all $z \in [l_i]$
3. From the Schwartz-Zippel lemma, it must hold that $\mathbf{C}^{(i+1)}[i_{1,z}, i_{2,z}] = \sum \hat{p}^{(i+1)}(X)\hat{p}_z(X) = \mathbf{y}[z]$ for all $z \in [l_i]$ with probability $1 - \frac{d_i l_i}{|\mathbb{F}|}$.

Based on the above, we first show that there exists a set of indexes $\mathcal{I}'_i \subseteq \mathcal{I}_i$ of size $|\mathcal{I}'_i| \geq l'_i$ such that $\Delta(\mathrm{Enc}_{\mathrm{RS}_i}(\mathbf{C}^{(i)}[i_1, i_2]), \mathbf{C}'^{(i)}[i_1, i_2]) < \frac{\delta_{\mathrm{RS}_i}}{2}$ for all $(i_1, i_2) \in \mathcal{I}'_i$. To prove this, assume that $|\mathcal{I}'_i| < l'_i$. Then, for the remaining indexes $\mathcal{I} \backslash \mathcal{I}'_i$ that correspond to segments the encodings of which are $\delta_{\mathrm{RS}_i}/2$-far, it must hold that $|\mathcal{I} \backslash \mathcal{I}'_i| \geq l_i - l'_i$. In this case, the verifier accepts with probability at most $(1 - \delta_{\mathrm{RS}_i}/2)^{l_i - l'_i}$ which however is negligible due to the fact that $l_i = \max(\frac{\lambda}{-\log(1-\delta_i/4)}, \frac{\lambda}{-\log(1-\delta_{\mathrm{RS}_i}/2)} + \frac{\lambda}{-\log(1-\delta_{\mathrm{RS}_i^{\otimes 2}}/2)})$. Thus with overwhelming probability, it must hold that $|\mathcal{I}'_i| \geq l'_i$.

Next, let $\mathrm{RS}'^{\otimes 2}_i = \mathrm{RS}^{\otimes 2}[(d, m_1^{(i)} m_2^{(i)}/d_i), (N, M_1^{(i)} m_2^{(i)}/d_i), \mathbb{F}]$. We will show that with overwhelming probability, it must hold that, $\Delta(\mathbf{C}^{(i)}, \mathrm{RS}_i^{\otimes 2}) < \delta_{\mathrm{RS}_i^{\otimes 2}}/2$. Assume that $\mathbf{C}^{(i)}$ is $\delta_{\mathrm{RS}_i^{\otimes 2}}/2$-far from $\mathrm{RS}_i^{\otimes 2}$. From Lemma 4 of Appendix 2 of [12] (but properly adapted for $\delta_{\mathrm{RS}^{\otimes 2}}/2$), however, we know that $\Delta(\mathrm{Fold}_{d_i}(\mathbf{C}^{(i)}, r_i), \mathrm{RS}'^{\otimes 2}_i) < \delta_{\mathrm{RS}'^{\otimes 2}_i}/2$ only with probability ϵ_{prox}, and thus, $\mathrm{Fold}_{d_i}(\mathbf{C}^{(i)}, r_i)$ would be far from the unique decoding radius of $\mathrm{RS}'^{\otimes 2}_i$. In other words, for random $\mathcal{I}'_i$ the probability that $\mathrm{Fold}_{d_i}(\mathbf{C}^{(i)}, r_i)$ agrees with $\mathrm{RS}'^{\otimes 2}_i$ is only $(1 - \frac{\delta_{\mathrm{RS}_i^{\otimes 2}}}{2})^{l'_i}$ which is negligible. In that case, conditions (1) and (3) can only hold with probability at most $(1 - \frac{\delta_{\mathrm{RS}_i^{\otimes 2}}}{2})^{l'_i} + \epsilon_{\mathrm{prox}} < \mathrm{negl}(\lambda)$. Consequently, with overwhelming

probability, $\mathbf{C}^{(i)}$ must be $\delta_{\mathrm{RS}_i^{\otimes 2}}/2$-close. Furthermore, because of the mutual correlated agreement, it must hold that $\mathrm{Enc}_{\mathrm{RS}^{\otimes 2}}(p^{(i)})[i_1, i_2] = \mathbf{C}^{(i)}[i_1, i_2]$ for all $(i_1, i_2) \in \mathcal{I}'_i$.

To this end, we proved that if the verifier passes the delegation checks, then, there must exist a query set $\mathcal{I}'_i \subseteq \mathcal{I}_i$ of size at least l'_i, such that $\mathcal{I}'_i \subset \mathcal{A}$. What remains is to show that $|\mathcal{A}| > (1 - \frac{\delta_{\mathrm{RS}_i^{\otimes 2}}}{2})N \cdot M_1$. Assume towards contradiction that $|\mathcal{A}| \leq (1 - \frac{\delta_{\mathrm{RS}_i^{\otimes 2}}}{2})N \cdot M_1$. Since $\mathcal{I}_i$ has been sampled after $[\mathbf{C}'^{(i+1)}]$ is fixed, the probability that the verifier accepts equals to the probability it samples at least l'_i segments in $\mathcal{A}$ which should be $(1 - \delta_{\mathrm{RS}^{\otimes 2}}/2)^{l'_i}$ and thus negligible. Consequently, $|\mathcal{A}| > (1 - \frac{\delta_{\mathrm{RS}_i^{\otimes 2}}}{2})N \cdot M_1$ with probability $1 - (1 - \delta_{\mathrm{RS}^{\otimes 2}}/2)^{l'_i}$. □

4.3.1 Round-by-Round Soundness.

In this section, we prove the round-by-round soundness of our protocol and in the next section round-by-round knowledge soundness. To begin with, assume that the prover starts in a doomed state, that is, it either holds that (1) $\Delta(\mathbf{C}^{(1)}, \mathrm{RS}_1^{\otimes 2,\mathrm{b}+1}) \geq \frac{\delta_{\mathrm{RS}_1^{\otimes 2}}}{2}$ or (2) $\Delta(\mathbf{C}^{(1)}, \mathrm{RS}_1^{\otimes 2,\mathrm{b}+1}) < \frac{\delta_{\mathrm{RS}_1^{\otimes 2}}}{2}$ but $K \neq \sum_{\omega_1,\omega_2 \in \mathbb{H}_1 \times \mathbb{H}_2} p(\omega_1, \omega_2) p_{\mathbf{b}}(\omega_1, \omega_2)$. In other words, we assume that $\mathbf{tr}_0 \in \mathcal{D}$ where $\mathbf{tr}_0 = (K, K_r, [\mathbf{C}_1], C_{p_r})$. We will show that for each step of the protocol, the prover can leave the doomed state only with negligible probability.

Masking Step (step 1): Conditioned that $\mathbf{tr}_0 \in \mathcal{D}$, then we will show that $\Pr[\mathbf{tr}_0 \notin \mathcal{D}] = \frac{1}{|\mathbb{F}|}$, where $\mathbf{tr}_1 = (\mathbf{tr}_0, r)$. That stems from the fact that since $\mathbf{tr}_0 \in \mathcal{D}$, then, either (1) $\Delta(\mathbf{C}, \mathrm{RS}_1^{\otimes 2,b+1}) \geq \frac{\delta_{\mathrm{RS}_1^{\otimes 2}}}{2}$ and thus $\Pr[\mathbf{tr}_1 \notin \mathcal{D}] = 0$ or (2) $\Delta(\mathbf{C}, \mathrm{RS}_1^{\otimes 2,b+1}) < \frac{\delta_{\mathrm{RS}_1^{\otimes 2}}}{2}$ but K or K_r are incorrect. In this case, the probability that the prover moves to an accepting state, is the probability that $K + rK_r = \sum p(x, y) p_{\mathbf{b}}(x, y) + r \sum p_r(x, y)$ which is only $\frac{1}{|\mathbb{F}|}$ due to the Schwartz-Zippel lemma.

Batching Step (steps 2, 3): Let $\mathbf{tr}_2 = (\mathbf{tr}_1, h_0(X_0), r_0, [\mathbf{C}^{(1)}])$ be the new transcript. Conditioned $\mathbf{tr}_1 \in \mathcal{D}$, we will show that $\Pr[\mathbf{tr}_2 \notin \mathcal{D}] \leq \max(\epsilon_{\mathrm{prox}}, \frac{2(d_0)}{|\mathbb{F}|})$— where ϵ_{prox} is the proximity error of RS tensor-codes. Since $\mathbf{tr}_1 \in \mathcal{D}$, it implies that (1) $\Delta(\mathbf{C}, \mathrm{RS}_1^{\otimes 2,\mathrm{b}+1}) \geq \frac{\delta_{\mathrm{RS}_1^{\otimes 2}}}{2}$ or (2) $\Delta(\mathbf{C}, \mathrm{RS}_1^{\otimes 2,\mathrm{b}+1}) < \frac{\delta_{\mathrm{RS}_1^{\otimes 2}}}{2}$ but K is incorrect. If case (2) holds, then the prover stays in a doomed state with probability $\frac{2d_0}{|\mathbb{F}|}$ due to the round-by-round soundness of our coSumcheck protocol. If case (1) holds, then, the prover has two options: either send $[\mathbf{C}^{(1)}]$ such that $\Delta(\mathbf{C}^{(1)}, \mathbf{RS}_1^{\otimes 2}) < \delta_{\mathrm{RS}_1^{\otimes 2}}/2$ or honestly follow the protocol. If it follows the first option it remains in a doomed state with probability 1. Looking ahead, this stems from the fact that the aggregation of $\mathbf{C}_1, \ldots, \mathbf{C}_{b+1}$ will differ significantly from $\mathrm{RS}_1^{\otimes 2}$ which will lead the verifier to reject in subsequent steps with overwhelming probability. Regarding the second option, the probability for an honestly computed $\mathbf{C}^{(1)}$ to be $\frac{\delta_{\mathrm{RS}_1^{\otimes 2}}}{2}$-close to a tensor-code is ϵ_{prox}.

Reduction Round (step 4, round $i = 1$): Let $\mathbf{tr}_3$ be the transcript produced by running the first round of step 3. It is parsed as $\mathbf{tr}_3 = (\mathbf{tr}_2, \mathbf{tr}_3^{(1)} = (h_i(X_i), r_i), \mathbf{tr}_3^{(2)} = ([\mathbf{C}'^{(2)}], \mathcal{I}_i = \{(i_{1,z}, i_{2,z}, i_{3,z})\}_{z\in[l_1]}, s), K^{(2)})$. We will show that if $\mathbf{tr}_2 \in \mathcal{D}$ then $\Pr[\mathbf{tr}_3 \notin \mathcal{D}] \leq \max(\frac{d_1}{|\mathbb{F}|}, (1 - \delta_{\mathrm{RS}_1^{\otimes 2}}/2)^{l_1})$.

1. $\mathbf{tr}_2 \to (\mathbf{tr}_2, \mathbf{tr}_3^{(1)})$: Assume that $\mathbf{tr}_2 \in \mathcal{D}$, that is, either (1) $\mathbf{C}^{(1)}$ is $\frac{\delta_{\mathrm{RS}_1^{\otimes 2}}}{2}$-far from $\mathrm{RS}_1^{\otimes 2}$ or inconsistent or (2) $\mathbf{C}^{(1)}$ is $\frac{\delta_{\mathrm{RS}_1^{\otimes 2}}}{2}$-close but $K^{(1)}$ is incorrect. Then, the probability $(\mathbf{tr}_2, \mathbf{tr}_3^{(1)}) \notin \mathcal{D}$ is $\frac{2d_1}{|\mathbb{F}|}$. That is because if (1) holds, then the prover remains in a doomed state with probability 1 while if (2) holds, then, the probability that $K^{(1)} \neq \sum \hat{p}^{(1)}(X_i, \mathbf{w})\hat{p}^{(1)}_{\mathbf{b}}(X_i, \mathbf{w})$ but $h_1(r_1) = \sum \hat{p}^{(1)}(r_1, \mathbf{w})\hat{p}^{(1)}_{\mathbf{b}}(r_1, \mathbf{w})$ is $2d_i/|\mathbb{F}|$ due to the round-by-round soundness of our sumcheck protocol.
2. $(\mathbf{tr}_2, \mathbf{tr}_3^{(1)}) \to (\mathbf{tr}_2, \mathbf{tr}_3^{(1)}, \mathbf{tr}_3^{(2)})$: Since $(\mathbf{tr}_2, \mathbf{tr}_3^{(1)}) \in \mathcal{D}$, then it must either hold that (1) $\mathbf{C}^{(1)}$ is $\frac{\delta_{\mathrm{RS}_1^{\otimes 2}}}{2}$-far from $\mathrm{RS}_1^{\otimes 2}$ (2) $\mathbf{C}^{(1)}$ is incorrectly computed or (3) $\mathbf{C}^{(1)}$ is $\frac{\delta_{\mathrm{RS}_1^{\otimes 2}}}{2}$-close but $h_1(r_1) \neq \sum \hat{p}^{(1)}(r_1, \mathbf{w})\hat{p}^{(1)}_{\mathbf{b}}(r_1, \mathbf{w})$. Furthermore, the new transcript moves to an accepting state if and only if $\mathbf{C}'^{(2)} \in \mathrm{Bind}_{p^{(2)}}$, $K^{(2)}$ is correct and all verification checks passed. We will show that this can happen only with probability $\frac{l_1}{|\mathbb{F}|} + (1 - \frac{\delta_{\mathrm{RS}_1^{\otimes 2}}}{2})^{l_1} + \epsilon_{\mathrm{prox}}$.
   (a) If case (1) holds, then, the probability that the prover leaves the doomed state equals to the probability that (i) $p^{(2)}(\omega_{i_{1,z}}, \omega_{i_{2,z}}) = \mathrm{Enc}_{\mathrm{RS}_1'^{\otimes 2}}(p^{(2)})[i_{1,z}, i_{2,z}] = \mathbf{y}[z]$ for all $(i_{1,z}, i_{2,z}) \in \mathcal{I}_1$ or (ii) there exist $\mathbf{y}[z] \neq p^{(2)}(\omega_{i_{1,z}}, \omega_{i_{2,z}})$ but $\sum_z s^z p^{(2)}(\omega_{i_{1,z}}, \omega_{i_{2,z}}) = \sum_z s^z \mathbf{y}[z]$. Since the last one happens only with probability $\frac{l_1}{|\mathbb{F}|}$ due to the Schwartz-Zippel Lemma, we focus on the former case. There, the prover might get "lucky" and $\mathrm{Fold}_{d_1}(\mathbf{C}^{(1)}, r_1))$ is $\delta_{\mathrm{RS}_1^{\otimes 2}}/2$-close, which can happen only with probability ϵ_{prox} or otherwise (i) holds only with probability $(1 - \delta_{\mathrm{RS}_1^{\otimes 2}}/2)^{l_1}$ for random $\mathcal{I}_1$.
   (b) If case (2) holds, then, the verifier can leave the doomed state if one of the following hold: (i) $\sum_z s^z p^{(2)}(\omega_{i_{1,z}}, \omega_{i_{2,z}}) = \sum_z s^z \mathbf{y}[z]$ but there exists $y[z] \neq p^{(2)}(\omega_{i_{1,z}}, \omega_{i_{2,z}})$ or (ii) $\mathbf{y}[z] = p^{(2)}(\omega_{i_{1,z}}, \omega_{i_{2,z}})$ for all z. As above, case (i) happens with probability at most $\frac{l_1}{|\mathbb{F}|}$ so we focus on (ii). First, assume that $\hat{p}^{(2)}(X) \neq \hat{p}^{(1)}(r_1, X)$. Then, $\Delta(\mathrm{Enc}(p^{(2)}), \mathrm{Fold}_{d_1}(\mathbf{C}^{(1)}, r_1)) > \delta_{\mathrm{RS}_1^{\otimes 2}}/2$, so, the probability the above is true for random $\mathcal{I}_1$ is $(1 - \delta_{\mathrm{RS}^{\otimes 2}}/2)^{l_1}$. Hence, we assume that $\hat{p}^{(2)}(X) = \hat{p}^{(1)}(r_1, X)$. In this case, due to the mutual correlated agreement, it must hold that $\mathbf{C}^{(1)}[i_{1,z}, i_{2,z}] = \mathrm{Enc}(p^{(1)})[i_{1,z}, i_{2,z}]$. Note however, that since $\Delta(\sum_{j\in[b+1]} \mathbf{b}_1[j]\mathbf{C}_j, \mathrm{RS}_1^{\otimes 2}) > \delta_{\mathrm{RS}_1^{\otimes 2}}/2$, it must hold that $\mathbf{C}^{(1)}[i_{1,z}, i_{2,z}][i_{3,z}] = \sum_{j\in[b+1]} \mathbf{b}_1[j]\mathbf{C}_j[i_{1,z}, i_{2,z}][i_{3,z}]$ for all $(i_{1,z}, i_{2,z})$ with probability $(1 - \delta_{\mathrm{RS}_1^{\otimes 2}}/2)^{l_1}$.

(c) In case (3), the prover leaves the doomed state if $K^{(2)}$ ends up being valid. For this the prover either can send $\mathbf{C}'^{(2)}$ such that $p^{(2)}(X) \neq p^{(1)}(r_1, X)$ but still the $K^{(2)}$ is valid, which as mentioned above can happen only with probability $\frac{l_1}{|\mathbb{F}|}$ or $p^{(2)}(X) = p^{(1)}(r_1, X)$ where in this case the prover stays in a doomed state with probability 1.

Reduction Round (step 4, $i > 1$ rounds): Let $\mathbf{tr}_{i+1}$ be the transcript produced by running the i-th round of step 3. It is parsed as $\mathbf{tr}_{i+1} = (\mathbf{tr}_i, \mathbf{tr}_{i+1}^{(1)} = (h_i(X_i), r_i), \mathbf{tr}_{i+1}^{(2)} = ([\mathbf{C}'^{(i+1)}], \mathcal{I}_i), \mathbf{tr}_{i+1}^{(3)} = ([\mathbf{R}], \mathcal{I}'_i, s), K^{(i+1)})$. We will show that for each $\mathbf{tr}_{i+1}^{(j)}$ the prover can move from a doomed state to a non-doomed one only with negligible probability.

1. $\mathbf{tr}_i \to (\mathbf{tr}_i, \mathbf{tr}_{i+1}^{(1)})$: Assume that $\mathbf{tr}_i \in \mathcal{D}$, that is, either (1) $\mathbf{C}'^{(i)} \notin$ Bind or (2) $\mathbf{C}'^{(i)} \in$ Bind but $K^{(i)}$ is incorrect. Then, the probability $(\mathbf{tr}_i, \mathbf{tr}_{i+1}^{(1)}) \notin \mathcal{D}$ is $\frac{2d_1}{|\mathbb{F}|}$. That is because if (1) holds, then the prover remains in a doomed state with probability 1 while if (2) holds, then, the probability that $K^{(i)} \neq \sum \hat{p}^{(i)}(X_i, \mathbf{w})\hat{p}_{\mathbf{b}}^{(i)}(X_i, \mathbf{w})$ but $h_i(r_i) = \sum \hat{p}^{(i)}(r_i, \mathbf{w})\hat{p}_{\mathbf{b}}^{(i)}(r_i, \mathbf{w})$ is $2d_i/|\mathbb{F}|$ due to the round-by-round soundness of our sumcheck protocol.

2. $(\mathbf{tr}_i, \mathbf{tr}_{i+1}^{(1)}) \to (\mathbf{tr}_i, \mathbf{tr}_{i+1}^{(1)}, \mathbf{tr}_{i+1}^{(2)})$: Since $(\mathbf{tr}_i, \mathbf{tr}_{i+1}^{(1)}) \in \mathcal{D}$, then it must either hold that (1) $\mathbf{C}'^{(i)} \notin$ Bind or (2) $\mathbf{C}'^{(i)} \in$ Bind but $h_i(r_i)$ is incorrect. Then, the probability $(\mathbf{tr}_i, \mathbf{tr}_{i+1}^{(1)}, \mathbf{tr}_{i+1}^{(2)}) \notin \mathcal{D}$ is at most $(1 - \delta_{\mathrm{RS}_i^{\otimes 2}}/2)^{l'_i}$. First, we assume that $\mathbf{C}'^{(i+1)} \in$ Bind and that $h_i(r_i) = \sum \hat{p}^{(i+1)}(\mathbf{w})\hat{p}_b^{(i+1)}(\mathbf{w})$ since otherwise the prover remains in a doomed state with probability 1. Next, we consider two cases:

   (a) Case 1: $\mathbf{C}'^{(i)} \notin$ Bind. In this case, the prover leaves the doomed state, if there exists $\mathbf{C}^{(i)}$ and at least l'_i indexes $(i_1, i_2) \in \mathcal{I}_i$, such that (1) $p^{(i+1)}(\omega_{1,i_1}, \omega_{2,i_2}) = \mathrm{Fold}_{d_i}(\mathbf{C}^{(i)}[i_1, i_2], r_i)$ and (2) $\mathbf{C}'^{(i)}[i_1, i_2]$ is $\delta_{\mathrm{RS}_i}/2$-close to the encoding of $\mathbf{C}^{(i)}[i_1, i_2]$. For a randomly chosen $\mathcal{I}_i$, however, this can happen only with probability at most $(1 - \delta_{\mathrm{RS}_i^{\otimes 2}}/2)^{l'_i}$ since $\mathbf{C}'^{(i)}$ must contain at most a fraction of $1 - \delta_{\mathrm{RS}_i^{\otimes 2}}/2$ such segments.

   (b) Case 2: $\mathbf{C}^{(i)} \in$ Bind then it must hold that $p^{(i)}(r_i, \vec{x}) \neq p^{(i+1)}(\vec{x})$. In this case, the prover leaves the doomed state if for at least l'_i indexes $(i_1, i_2) \in \mathcal{I}_i$, it must hold that $p^{(i+1)}(\omega_{1,i_1}, \omega_{2,i_2}) = \mathrm{Fold}_{d_i}(\mathbf{C}'^{(i)}[i_1, i_2], r_i))$. However, the chance of this happening for random $\mathcal{I}_i$ is $(1-\delta_{\mathrm{RS}_i^{\otimes 2}}/2)^{l'_i}$.

That is because:

$$\Delta(\mathrm{Enc}_{\mathrm{RS}_i'^{\otimes 2}}(p^{(i+1)}), \mathrm{Fold}(\mathrm{Enc}_{\mathrm{RS}_i^{\otimes 2}}(p^{(i)}), r_i))) > \delta_{\mathrm{RS}_i^{\otimes 2}} \equiv$$
$$\Delta(\mathrm{Enc}_{\mathrm{RS}_i'^{\otimes 2}}(p^{(i+1)}), \mathrm{Fold}(\mathbf{C}'^{(i)}, r_i)) + \Delta(\mathrm{Fold}(\mathbf{C}'^{(i)}, r_i), \mathrm{Fold}(\mathrm{Enc}_{\mathrm{RS}_i^{\otimes 2}}(p^{(i)}), r_i))) > \delta_{\mathrm{RS}_i^{\otimes 2}} \equiv$$
$$\Delta(\mathrm{Enc}_{\mathrm{RS}_i'^{\otimes 2}}(p^{(i+1)}), \mathrm{Fold}(\mathbf{C}'^{(i)}, r_i)) + \delta_{\mathrm{RS}_i^{\otimes 2}}/2 > \delta_{\mathrm{RS}_i^{\otimes 2}} \equiv$$
$$\Delta(\mathrm{Enc}_{\mathrm{RS}_i'^{\otimes 2}}(p^{(i+1)}), \mathrm{Fold}(\mathbf{C}'^{(i)}, r_i)) > \delta_{\mathrm{RS}_i^{\otimes 2}}/2$$

3. $(\mathbf{tr}_i, \mathbf{tr}_{i+1}^{(1)}, \mathbf{tr}_{i+1}^{(2)}) \rightarrow (\mathbf{tr}_i, \mathbf{tr}_{i+1}^{(1)}, \mathbf{tr}_{i+1}^{(2)}, \mathbf{tr}_{i+1}^{(3)})$: Since $(\mathbf{tr}_i, \mathbf{tr}_{i+1}^{(1)}, \mathbf{tr}_{i+1}^{(2)}) \in \mathcal{D}$, it will still hold that (1) $\mathbf{C}^{(i)} \notin$ Bind or (2) $\mathbf{C}^{(i)} \in$ Bind but $h_i(r_i) \neq \sum \hat{p}^{(i)}(r_i, \mathbf{w}) \hat{p}_{\mathbf{b}}^{(i)}(r_i, \mathbf{w})$. The prover leaves the doomed state if all verification constraints hold and $\mathbf{C}^{(i+1)} \in$ Bind and $K^{(i+1)} = \sum p^{(i+1)}(\vec{\omega}) p_{\mathbf{b}}^{(i+1)}(\vec{\omega})$. This, however, can happen only with probability ϵ_{bind}. In particular, if case (1) holds, then, by Lemma 5 the verifier should pass all checks with probability at most ϵ_{bind}. Next, if case (2) holds, and because the original state is doomed, there should exist $(i_1, i_2) \in \mathcal{I}_i$ such that $p^{(i+1)}(\omega_{1,i_1}, \omega_{2,i_2}) \neq \mathrm{Fold}_{d_i}(\mathbf{C}'^{(i)}[i_1, i_2], r_i)$ and hence, due to the Schwartz-Zippel lemma, $K^{(i+1)} = \sum p^{(i+1)}(\mathbf{w}) p_{\mathbf{b}}^{(i+1)}(\mathbf{w})$ only with probability $l_i/|\mathbb{F}|$ for a random challenge $s \in \mathbb{F}$.

4.3.2 Round-by-Round Knowledge Soundness.

Initially, assume that the initial instance belongs to $\mathcal{R}_{ip}$, namely $\Delta(\mathbf{C}, \mathrm{RS}_1^{\otimes 2, b+1}) < \frac{\delta_{\mathrm{RS}^{\otimes 2}}}{2}$ and $K = \sum p(\omega_1, \omega_2) p_{\mathbf{b}}(\omega_1, \omega_2)$. Then we can construct an extractor which is given access to the prover and outputs p. Our extractor uses the straightline extractor of [13] to obtain $\mathbf{C}$. Since, $\mathbf{C}$ is $\delta_{\mathrm{RS}_1^{\otimes 2, b+1}}/2$ close, there will exist a unique $p : \mathbb{F}^2 \rightarrow \mathbb{F}$ which can be obtained by decoding $\mathbf{C}_i$ for all $i \in [b+1]$ using the algorithm proposed in Section 1.2.

Having established that we can now show round-by-round knowledge. In particular, we will show that if $\Pr[\mathbf{tr}_{i+1} \notin \mathcal{D}] > \epsilon_{\mathrm{rbr}}$, then, it must hold that $\mathbf{tr}_i \notin \mathcal{D}$ for all transcripts $\mathbf{tr}_i, \mathbf{tr}_{i+1}$ which correspond to the i-th and $(i+1)$-th step of the protocol respectively. This would imply that $\mathbf{tr}_0 \notin \mathcal{D}$ and hence the extractor would output a valid witness.

Masking Step (step 1): Assume that $\mathbf{tr}_0 \in \mathcal{D}$ and suppose that $\Pr[\mathbf{tr}_1 = (\mathbf{tr}_0, r) \notin \mathcal{D}] > \epsilon_{\mathrm{rbr}}$. We will show that $\mathbf{tr}_0 \notin \mathcal{D}$. For this, assume that $\mathbf{tr}_0 \in \mathcal{D}$. Recall that the probability $\mathbf{tr}_1 \notin \mathcal{D}$ conditioned that $\mathbf{tr}_0 \in \mathcal{D}$ is at most $1/|\mathbb{F}|$ which contradicts our initial assumption since $\frac{1}{|\mathbb{F}|} < \epsilon_{\mathrm{rbr}}$. Hence $\mathbf{tr}_0 \notin \mathcal{D}$ and thus, given $\mathbf{tr}_1$, the extractor can output a valid witness.

Batching Step (steps 2, 3). Assume that $\mathbf{tr}_1 \in \mathcal{D}$ and suppose that $\Pr[\mathbf{tr}_2 = (\mathbf{tr}_0, \mathbf{C}^{(1)}) \notin \mathcal{D}] > \epsilon_{\mathrm{rbr}}$ for any $\mathbf{C}^{(1)}$. We will end up in a contradiction. Since $\mathbf{tr}_2 \notin \mathcal{D}$, it must hold that $\mathbf{C}^{(1)}$ is computed honestly, it is $\delta_{\mathrm{RS}_1^{\otimes 2}}/2$-close to $\mathrm{Enc}_{\mathrm{RS}_1^{\otimes 2}}(p^{(1)})$ and the instance $K^{(1)}$ is correct. As we shown in the previous section however, this can happen only with probability at $\max(\epsilon_{\mathrm{prox}}, \frac{2d_0}{|\mathbb{F}|}) < \epsilon_{\mathrm{rbr}}$

which contradicts the assumption that $\mathbf{tr}_1 \in \mathcal{D}$. Consequently, it must hold that $\mathbf{tr}_1 \notin \mathcal{D}$ and thus given $\mathbf{tr}_2$, the extractor can output a valid witness.

Reduction Round (step 4, i-th round): Assume that $\Pr[\mathbf{tr}_{i+1} \notin \mathcal{D}] > \epsilon_{\text{rbr}}$. We will show that this holds if and only if $\mathbf{tr}_i \notin \mathcal{D}$. Assume towards contradiction that $\mathbf{tr}_i \notin \mathcal{D}$. In this case, however, the probability that $\mathbf{tr}_{i+1}$ moves to a non-doomed state is at most $(1 - \delta_{\text{RS}_i^{\otimes 2}}/2)^{l'_i} + \epsilon_{\text{bind}} + \frac{2d_1}{|\mathbb{F}|} \leq \epsilon_{\text{rbr}}$ which contradicts our initial assumption.

4.4 t-Zero Knowledge

Theorem 8. ***Protocol 1*** *is honest-verifier t-zero knowledge in the $(\Pi^{Enc}_{RS_1}, \Pi^{Sumcheck})$-hybrid model.*

Proof. Let $\mathcal{A}$ be an adversary that controls at most t provers $\mathcal{P}_{i_j}$ for all $i_j \in \mathcal{C}$ as well as the verifier. Observe that the view of $\mathcal{A}$ is equivalent to $\text{View}_{\mathcal{A}}(K, \mathbf{X})$ (1) the oracles of all codewords $[\mathbf{C}], \{[\mathbf{C}^{(i)}]\}_{i\in[k]}$, (2) the codeword rows $\{\mathbf{C}_{i,j}\}_{i\in\mathcal{C}, j\in[b+1]}$, held by the corrupted provers, (3) the query replies $\mathbf{C}[\mathcal{I}'_1 \times [b+1]]$, $\{\mathbf{C}^{(i)}[\mathcal{I}_i]$ and, finally, (4) the coSumcheck replies $h^{(i)}\}_{i\in\{0,k-1\}}, \hat{p}_r^{(k)}, \{\hat{p}_i\}_{i\in\mathcal{C}}$ and $\{K_{i,r}\}_{i\in[d]}$. Note that the view does not contain the rows of the folded codewords of the corrupted provers since they are computed deterministically. Furthermore, recall that $\mathcal{I}'_1 \times [b+1]$ are the query indexes sampled by the verifier at step $4.b.i$ and they used to validate the correctness of the batching where $\mathcal{I}'_1 \subset \mathcal{I}_1$.

Before delving into the construction of the simulator, we first make some observations regarding the distribution of $\text{View}_{\mathcal{A}}$. In particular, we will show that $\text{View}_{\mathcal{A}}$ follows a uniform distribution (over accepting proofs). In other words, given any $\mathbf{X}$ such that $K = \sum_{i,j} \mathbf{X}[i,j]\mathbf{B}[i,j]$, we will show how to use (1)-(3) to find the padded by randomness matrix $\mathbf{X}^*$ and then we will use it along with (4) to find the unique masking polynomial p_r.

Finding $\mathbf{X}^*$. First, using $\{\mathbf{C}_{i,j}\}_{i\in[b],j\in\mathcal{C}}$, $\mathbf{X}$ and $\mathbf{C}[\mathcal{I}'_1 \times [b]]$, one can find the unique sub-matrices $\{\mathbf{X}^*_i\}_{i\in[b]}$ such that $\mathbf{C}_i = \text{Enc}_{\text{RS}_1^{\otimes 2}}(\mathbf{X}^*)$. This directly stems from the (t, l_1)-zero-knowledge property of $\mathbf{C}_i$ for all $i \in [b]$. What remains, is to find the randomized matrix $\mathbf{X}^*_{b+1}$ which is done by first constructing the matrix $\mathbf{X}^{*(1)}$ that encodes the (t, l)-zero-knowledge codeword $\mathbf{C}^{(1)}$. In more detail, we set $\mathbf{X}^{*(1)}[i] = \sum_{j\in[b]} L_{d_0,j}(r_0)\mathbf{X}^*_j[i]$, in all non-random columns and then using $\{\mathbf{C}^{(i)}[\mathcal{I}_i]\}_{i\in[k]}$ and the algorithm of Lemma 3, we obtain a unique $\mathbf{X}^{*(1)}$ with probability at least $1 - \epsilon_{\text{zk,whir}}$. Finally, we set $\mathbf{X}^*_{b+1} = L_{d_0,b+1}(r_0)^{-1}(\mathbf{X}^{*(1)} - \sum_{j\in[b]} L_{d_0,j}(r_0)\mathbf{X}^*_j)$. Note that since $\mathcal{I}'_1 \subset \mathcal{I}_1$, there are no additional constraints to be enforced. Finally note that the above algorithm fails with an additional probability of $\frac{d_0}{|\mathbb{F}|}$ e.g., $L_{d_0,b+1}(r_0) = 0$.

Finding $\hat{p}_r$. Given $\mathbf{X}^*$, and (4) we can now find a unique masking polynomial by following the same algorithm as in Theorem 3.1.

Based on the above, we conclude that $\text{View}_{\mathcal{A}}(K, K_r, \mathbf{X})$ follows a uniform distribution with probability $1 - \epsilon_{\text{zk,whir}} - \epsilon_{\text{sck}} > 1 - \text{negl}(\lambda)$—where ϵ_{sck} the

statistical error of the sumcheck protocol. To prove t-zero-knowledge of **Protocol 1** it is enough to construct a simulator that produces a view following a uniform distribution. Our simulator works as follows:

1. The simulator first calls simulator of Theorem 6 to produce the views $(\{\mathbf{C}_i^{(1)}\}_{i\in\mathcal{C}}, \{\mathbf{C}^{(i)}[\mathcal{I}_i]\}_{i\in[k]})$ and then the simulator of the coSumcheck protocol to obtain $(\{h^{(i)}\}_{i\in[k]}, K_{r,i}, \{\hat{p}_i\}_{i\in\mathcal{C}})$.

2. It samples the codewords $\{\mathbf{C}_{i,j}\}_{i\in[b],j\in\mathcal{C}}$ uniformly at random with the constraints that they are consistent with $\{\mathbf{X}_i\}_{i\in\mathcal{C}}$ and the random tape of each corrupted prover and evaluates $\mathbf{C}_{(b+1),i}$ using $\mathbf{C}_i^{(1)}$ and $\mathbf{C}_{j,i}$ for all $i \in \mathcal{C}$ and $j \in [b]$.

3. It randomly samples $\mathbf{C}[\mathcal{I}_1' \times [b+1]]$ such that $\sum_{j\in[b+1]} \mathbf{b}_1[j]\mathbf{C}[i,j] = \mathbf{C}^{(1)}[i]$ for all $i \in \mathcal{I}_1' \subset \mathcal{I}_1$.

4. Initializes the oracles $[\mathbf{C}], \{\mathbf{C}^{(i)}\}_{i\in[k]}$ to be consistent with the queried points and outputs $([\mathbf{C}], \{[\mathbf{C}^{(i)}]\}_{i\in[k]}, \{\mathbf{C}_{i,j}\}_{i\in\mathcal{C},j\in[b+1]}, \mathbf{C}[\mathcal{I}_1' \times [b+1]], \{\mathbf{C}^{(i)}[\mathcal{I}_i], h^{(i)}\}_{i\in[k]}, y_r)$

□

5 Proof of Theorem 9 (Our coPIOP)

In this section, we present our constant-round coPIOP for proving R1CS instances. In practice, our protocol is almost identical to that of [12] but instead of using multi-linear polynomials to encode the proving data, we use multivariate polynomials with a small number of variables. This enables us to also invoke our constant-round coSumcheck. For completeness we provide a detailed description of our constant round sumcheck protocol in **Protocol 3** and our coPIOP in **Protocol 4**.

Protocol 3: Constant round t-zero-knowledge collaborative sumcheck protocol for proving $K = \sum_{\omega\in\mathbb{H}_1\times\ldots\times\mathbb{H}_k} \hat{p}_1(\omega)\hat{p}_2(\omega)$, where $\hat{p}_1, \hat{p}_2 : \mathbb{F}^k \to \mathbb{F}$, k-variate polynomials of individual degrees $(d_1, \ldots, d_k)$ that interpolate the matrices $\mathbf{X}_1, \mathbf{X}_2 \in \mathbb{F}^{d\times m}$ where $d = d_k$ at $\mathbb{H}_1 \times \ldots \times \mathbb{H}_k$.

Pre-Processing. Let $g(x_1, \ldots, x_k)$ be the masking polynomial of the form $g(x_1, \ldots, x_k) = \sum_{i\in[k]} g^{(i)}(x_i) \prod_{j\in[k]\setminus i} L_{j,0}(x_j)$ where $L_{j,0}$ is the Lagrange polynomial of the x_j-th variable. Each $g^{(i)}$ has degree $2d_i$ and $g^{(i)}(\omega_{i,1}) = 0$ for each $i \in [k-1]$ and $g^{(k)}$ a random d_k-degree polynomial. In practice, g interpolates a matrix $\mathbf{R} \in \mathbb{F}^{d\times 2m}$ at $\mathbb{H}_1' \times \ldots \times \mathbb{H}_{k-1}' \times \mathbb{H}_k$ where $d = d_k$, $\mathbb{H}_i \subset \mathbb{H}_i'$ with $|\mathbb{H}_i'| = 2|\mathbb{H}_i|$ and all elements of $\mathbf{R}$ are zero except

from those at the points $(\omega_{1,1}, \ldots, \omega_{i-1,1}, \mathbb{H}'_i, \omega_{i+1,1}, \ldots, \omega_{k,1})$ which are selected uniformly at random. Provers invoke $\Pi^{\text{Rand},p}$ functionality to produce packed secret shares of g. In particular, $\Pi^{\text{Rand},p}$ samples the random matrix $\mathbf{R} \in \mathbb{F}^{d \times 2\sum_{i\in[k-1]} d_i}$ as described above and pads it with t random rows forming $\mathbf{R}^* \in \mathbb{F}^{(d+t)\times 2\sum_{i\in[k-1]} d_i}$. Finally it computes the packed shares of each column of $\mathbf{R}^*$ and sends the i-th share to $\mathcal{P}_i$. Each $\mathcal{P}_i$ ends up with $g_i(x_1, \ldots, x_{k-1}) = g^*(x_1, \ldots, x_{k-1}, \omega'_{i,k}) : \mathbb{F}^{k-1} \to \mathbb{F}$ where g^* the polynomial that interpolates $\mathbf{R}^*$. Finally, all parties invoke Π^{Sum} to compute $K_{i,r} = \sum_{\omega\in\mathbb{H}_1\times\ldots\mathbb{H}_{k-1}} g(\omega, \omega_{k,i})$ for all $i \in [d]$ as well as $K_r = \sum K_{i,r}$.

Prove Protocol. Each prover $\mathcal{P}_i$ is given g_i and also $p_{1,i}, p_{2,i} : \mathbb{F}^{k-1} \to \mathbb{F}$, the i-th secret shares of p_1 and p_2 respectively. The verifier holds polynomial oracles of p_1, p_2 and g respectively. Next, all parties interact over the following protocol:

1. $\mathcal{V}$: Sample a random point $a \in \mathbb{F}$ and send it to all provers. All parties set $K^{(1)} \leftarrow K + aK_r$. Furthermore, each prover $\mathcal{P}_j$ sets $p^{(1)}_{1,j} \leftarrow p_{1,j}$ and $p^{(1)}_{2,j} \leftarrow p_{2,j}$.
2. For $i = 1, \ldots, k-1$ repeat the following:
   (a) $\mathcal{P}_j$: Compute $h^{(j)}_i(x_i) = [\![h_i(x_i)]\!]_i = \sum_{\omega\in\mathbb{H}_{i+1}\times\ldots\times\mathbb{H}_{k-1}} p^{(i)}_{1,j}(x_i, \omega)p^{(i)}_{2,j}(x_i, \omega) + a \cdot g^{(i)}_j(x_i, \omega)$.
   (b) $\mathcal{P}_1$-$\mathcal{P}_N$: Invoke the $\Pi^{\text{PSS,ip}}$ functionality to compute $h_i(x_i) = \sum_{\omega_1\in\mathbb{H}_1} h_i(\omega_1, x_i)$ and send it to $\mathcal{V}$.
   (c) $\mathcal{V}$: Checks if $K^{(i)} = \sum_{\omega\in\mathbb{H}_i} h_i(\omega_i)$, samples a random point $r_i \in \mathbb{F}$ and sends it to the provers. All parties compute $K^{(i+1)} \leftarrow h_i(r_i)$.
   (d) $\mathcal{P}_j$: Set $p^{(i+1)}_{l,j}(x_{i+1}, \ldots, x_{k-1}) \leftarrow p^{(i)}_{l,j}(r_i, x_{i+1}, \ldots, x_{k-1})$ for $l = \{1, 2\}$ and $g^{(i+1)}_j(x_{i+1}, \ldots, x_{k-1}) \leftarrow g^{(i)}_j(r_i, x_{i+1}, \ldots, x_{k-1})$.
3. $\mathcal{P}_j$: Send $p_{1,j}(r_1, \ldots, r_{k-1}), p_{2,j}(r_1, \ldots, r_{k-1}), g_j(r_1, \ldots, r_{k-1})$ to $\mathcal{V}$.
4. $\mathcal{V}$: Compute $p_1(r_1, \ldots, r_{k-1}, x_k), p_2(r_1, \ldots, r_{k-1}, x_k)$ and $g(r_1, \ldots, r_{k-1}, x_k)$. Check if $K^{(k)} = \sum_{\omega\in\mathbb{H}_k} p_1(r_1, \ldots, r_{k-1}, \omega)p_2(r_1, \ldots, r_{k-1}, \omega) + ag(r_1, \ldots, r_{k-1}, \omega)$. Finally, sample a random point r_k and invoke the oracles of p_1, p_2 and g to validate the correctness of $p(r_1, \ldots, r_{k-1}, r_k)$ and $g(r_1, \ldots, r_{k-1}, r_k)$.

Protocol 4: Our Constant-Round coPIOP.

$\mathcal{P}_i$'s inputs. Each prover holds portion of the R1CS matrixes $(\mathbf{A}, \mathbf{B}, \mathbf{C})$, and the packed secret shares of the columns of: (1) the witness matrix $\mathbf{W}^* = (\mathbf{W}, \mathbf{W}_r = (\mathbf{r}, \mathbf{0}, \ldots, \mathbf{0})) \in \mathbb{F}^{d \times 2n/d}$, where $\mathbf{r} \in \mathbb{F}^d$ a random column and (2) $\mathbf{U}_L^* = (\mathbf{U}_L, (\mathbf{R}_1, \mathbf{0}, \ldots, \mathbf{0}))$, $\mathbf{U}_R^* = (\mathbf{U}_R, (\mathbf{R}_2, \mathbf{0}, \ldots, \mathbf{0}))$, $\mathbf{U}_O = (\mathbf{U}_O, (\mathbf{R}_3, \mathbf{0}, \ldots, \mathbf{0})) \in \mathbb{F}^{d \times 2m/d}$, where $\mathbf{R}_1, \mathbf{R}_2 \in \mathbb{F}^{d \times 2}$ and $\mathbf{R}_3 \in \mathbb{F}^{d \times 3}$ and $\mathbf{V}_L, \mathbf{V}_R, \mathbf{V}_O$ the matrices formed by the vectors $\mathbf{v}_L = \mathbf{A}\mathbf{w}$, $\mathbf{v}_R = \mathbf{B}\mathbf{w}$ and $\mathbf{v}_O = \mathbf{C}\mathbf{w}$ respectively. Note that the secret shares of the random vectors have been produced in a pre-processing phase. Using these shares, each prover $\mathcal{P}_i$ forms its secret-shared polynomials $p_{w,i} : \mathbb{F}^{k_1 - 1} \to \mathbb{F}$ and $p_{L,i}, p_{R,i}, p_{O,i} : \mathbb{F}^{k_2} \to \mathbb{F}$ of degrees $(d_{1,1}, \ldots, d_{1,k_1-1})$ and $(d_{2,1}, \ldots, d_{2,k_2-1} = 1)$ where p_w, p_L, p_R, p_O interpolates the padded by randomness matrices $\mathbf{W}^*$, $\mathbf{U}_L^*$, $\mathbf{U}_R^*$ and $\mathbf{U}_O^*$ respectively.

$\mathcal{V}$'s inputs. The verifier holds the polynomial oracles of the $(d_{1,1}, \ldots, d_{k_1-1}, d, d_{2,1}, \ldots, d_{2,k_2-2}, d)$-degree polynomials $p_A, p_B, p_C : \mathbb{F}^{k_1+k_2-1} \to \mathbb{F}$ that interpolate $\mathbf{A}, \mathbf{B}$ and $\mathbf{C}$.

Protocol Π^{coPIOP}. Each prover works as follows:

1. $\mathcal{P}_1, \ldots, \mathcal{P}_N$: Send the polynomial oracles of p_w, and $p_1, p_2, p_3 : \mathbb{F}^{k_1 - 1} \to \mathbb{F}$ which are the polynomials that interpolate $\mathbf{R}_1, \mathbf{R}_2$ and $\mathbf{R}_3$ respectively.

2. $\mathcal{V}$: Sample a random point $\mathbf{r} \in \mathbb{F}^{\log m}$ and send it to the provers.

3. $\mathcal{P}_1, \ldots, \mathcal{P}_N$: All provers invoke the Π^{ip} functionality to compute $K' = \sum_{i,j} \beta((\mathbf{i}, \mathbf{j}), \mathbf{r})\big(\mathbf{R}_1[i,j]\mathbf{R}_2[i,j] - \mathbf{R}_3[i,j]\big)$. Finally, $\mathcal{P}_1$ sends K' to $\mathcal{V}$ which replies with another random point r'.

4. $\mathcal{P}_1, \ldots, \mathcal{P}_N, \mathcal{V}$: Set $\mathbf{r} \leftarrow (\mathbf{r}, r')$ and interact following **Protocol 3** to prove that:

$$r'K' = \sum_{\omega \in \mathbb{H}_1 \times \ldots \times \mathbb{H}_{k_2}} p_\beta(\omega)\Big(p_L(\omega)p_R(\omega) - p_O(\omega)\Big)$$

   Where p_β the polynomial with the same degree as p_L, p_R, p_O that interpolates $(\beta(\mathbf{0}, \mathbf{r}), \ldots, \beta(\mathbf{2m\text{-}1}, \mathbf{r}))$. At the end of the protocol, parties end up with the polynomials $p_L(r'_1, \ldots, r'_{k_2-1}, x_{k_2})$, $p_R(r'_1, \ldots, r'_{k_2-1}, x_{k_2})$ and $p_O(r'_1, \ldots, r'_{k_2-1}, x_{k_2})$ as well as $y_L = p_L(\mathbf{r}')$, $y_R = p_R(\mathbf{r}')$, $y_O = p_O(\mathbf{r}')$ where $\mathbf{r}' = (r_1, \ldots, r_{k_2})$.

5. $\mathcal{V}$: Validates the correctness of $y_\beta = p_\beta(\mathbf{r}')$ locally, and sends the random points $a, b, c \in \mathbb{F}$.

6. $\mathcal{P}_1, \ldots, \mathcal{P}_N$: All provers collaborate to compute the packed shares of the vector $\mathbf{a}$, computed by evaluating the polynomial $p_{\text{R1CS}}(\mathbf{x}) = ap_A(\mathbf{r}', \mathbf{x}) + bp_B(\mathbf{r}', \mathbf{x}) + cp_C(\mathbf{r}', \mathbf{x})$ at all points $\mathbb{H}'_1 \times \ldots \times \mathbb{H}'_{k_1}$. Then

set $\mathbf{p}' = (\mathbf{p}, \mathbf{0}) \in \mathbb{F}^{2n}$. In addition, compute the secret shares of the vector $\mathbf{b} = [\beta_L((0,\mathbf{0}),\mathbf{r}'),\ldots,\beta_L((0,\mathbf{n\text{-}1}),\mathbf{r}')] \in \mathbb{F}^n$ where $\beta_L(\mathbf{x},\mathbf{y}) = \prod_{i\in[|\mathbf{y}|]} L_{i,x[i]}(y[i])$. Set $\mathbf{b}' = (\mathbf{b},\mathbf{0}) \in \mathbb{F}^{2n}$.

7. $\mathcal{P}_1,\ldots,\mathcal{P}_N,\mathcal{V}$: Interact following **Protocol 3** for proving the instance:

$$ay_L+by_R+cy_O = \sum_{\omega\in\mathbb{H}'_1\times\ldots\times\mathbb{H}'_{k_1}} \Big(p_{\mathrm{R1CS}}(\omega)p_w(\omega)+p_{\mathbf{b}'}(\omega)\big(ap_1(\omega)+bp_2(\omega)+cp_3(\omega)\big)\Big)$$

   At the end of the protocol, the verifier ends up with the evaluation claims y_w, y_A, y_B, y_C, $y_{b'}$ and y_1, y_2, y_3 of the polynomials $p_w, p_A, p_B, p_C, p_{\mathbf{b}'}, p_1, p_2, p_3$ at $\mathbf{r}'' = (r''_1,\ldots,r''_{k_1})$. Furthermore, parties also obtain $p_w(r''_1,\ldots,r''_{k_1-1},x_{k_1})$ as well as $p_i(r''_1,\ldots,r''_{k_1-1},x_{k_1})$ for all $i \in [3]$.

8. $\mathcal{V}$: Validate $y_{b'}$ locally and for the rest of the polynomials use their respective oracles.

Finally we can claim the following:

Theorem 9. ***Protocol 4*** *is a coPIOP with $\mathcal{O}(k_1+k_2)$ communication rounds among the provers, $\mathcal{O}(|C|\log|C|/(\frac{\epsilon}{2}N))$ prover complexity and $5N+\mathcal{O}(polylog(m))$ proof size and verification time. Furthermore, it has $\epsilon_{rbr} = \kappa_{rbr} = \max(\frac{\sum_i d_{1,i}}{|\mathbb{F}|}, \frac{\sum_i d_{2,i}}{|\mathbb{F}|}, (1-\delta)^l)$ round-by-round (knowledge) soundness, where δ is the proximity parameter used when instantiating the PC scheme that opens the masked polynomials, and t-zero-knowledge with a $\epsilon_{zk} = \frac{3(\sum d_{1,i}+\sum d_{2,i})}{|\mathbb{F}|}$ statistical error and $t \le (1-\epsilon)N/2$, where $\epsilon \in (0,1)$.*

Proof. The proofs for round-by-round (knowledge) soundness and completeness are identical to that of [12] but adapted to encompass the sumcheck of **Protocol 3**. In the subsequent sections, we will focus on t-Zero-Knowledge and performance analysis. □

Corollary 1. *By replacing the polynomial oracles with our coIOPP, we obtain a coIOPP for R1CS that is round-by-round (knowledge) sound and honest-verifier t-zero-knowledge (as per Theorem 6 of [12]). Finally, by applying the coBCS and because of Theorem 1, the final non-interactive protocol will be a coSNARK.*

5.1 Performance Analysis

Performance Analysis of our coSumcheck. We start with the performance analysis of the coSumcheck of **Protocol 3**. Let $M = d_1 \cdot \ldots \cdot d_k$ be the size of the polynomials. For each round i and for each prover, step 2.a requires

$\mathcal{O}((\frac{M}{d_k}\log d_i)/(\prod_{j\in[i-1]} d_j))$ field operations, step 2.b requires $\mathcal{O}(N+d_i)$ field operations while step 2.d requires $\mathcal{O}((\frac{M}{d_k})/(\prod_{j\in[i-1]} d_j))$ field operations. So, the total per-prover complexity is $\mathcal{O}(\sum_i(\frac{M}{d_k}\log d_i)/(\prod_{j\in[i-1]} d_j))$. If we set $d_1 \in \mathcal{O}(\log M)$, $k > 2$ and $d_2, \ldots, d_{k-1} \in \mathcal{O}(\sqrt[k-1]{M/d_k})$ (similarly to [11]), then, the total prover complexity is $\mathcal{O}(\frac{M}{N}\log\log M)$. Regarding the proof size, naively, it is $\mathcal{O}(\sum_i 2d_i)$ dominated by step 2.c and $3N$ field elements derived from step 3. As mentioned in the main body, we use the proof composition protocol of [11] to reduce that to only $3N + \mathcal{O}(\log^2 M)$ field elements. Likewise, we argue about verification time. Finally, the total communication cost is $\mathcal{O}(N\sum d_i) = \mathcal{O}(N\sqrt[k-1]{\frac{M}{N}})$ (derived from the instantiation of the batched $\Pi^{\text{PSS,ip}}$ of Section 6.1.1) and the number of communication rounds are $\mathcal{O}(k)$.

Performance Analysis of our coPIOP. Note that the per-prover complexity of our coPIOP is solely dominated by the invocations of the sumcheck protocol both of them require $\mathcal{O}(\frac{|C|}{N}\log\log|C|)$ field operations. As for the proof size (and likewise for verification time), it is solely derived by the sumcheck protocol, which is $\mathcal{O}(\log^2|C|) + 9N$ field elements. Finally, the protocol requires $\mathcal{O}(k_1 + k_2)$ communication rounds among the provers which is $\mathcal{O}(1)$ for k_1, k_2 small constants and the total communication complexity is $\mathcal{O}(|C|)$—dominated by step 6.

5.2 Proof of t-Zero-Knowledge of our Sumcheck

Here we show that our coSumcheck protocol presented in **Protocol 3** is (honest-verifier) t-zero-knowledge where $t + d \leq N/2$. In particular, we will prove the following theorem:

Theorem 10. *For any semi-honest adversary $\mathcal{A}$ corrupting at most $t \leq N/2-d$ provers $\{\mathcal{P}_i\}_{i\in[\mathcal{C}]}$, there exists a simulator $\mathcal{S}$ such that the following views are indistinguishable:*

1. *The view $View_{\mathcal{R},\mathcal{A}}(K, p_1(r_1,\ldots,r_{k-1},x_k), p_2(r_1,\ldots,r_{k-1},x_k))$ of the adversary when running **Protocol 3** in the real world.*
2. *The view $View_{\mathcal{S},\mathcal{A}}(K, p_1(r_1,\ldots,r_{k-1},x_k), p_2(r_1,\ldots,r_{k-1},x_k))$ produced by the simulator.*

As a first step, observe that the view of the adversary consists of:

1. The packed secret shares of corrupted provers $\{p_{1,j}, p_{2,j}, g_j\}_{j\in\mathcal{C}}$.
2. The sums $K_{1,r}$, …, $K_{d,r}$.
3. The sumcheck polynomials $h_i(x_i)$ for all $i \in [k-1]$.
4. The evaluations $\{p_{1,j}(r_1,\ldots,r_{k-1}), p_{2,j}(r_1,\ldots,r_{k-1}), g_j(r_1,\ldots,r_{k-1})\}_{j\in[N]\setminus\mathcal{C}}$.
5. The final polynomial $g(r_1,\ldots,r_{k-1},x_k)$.

Note that given (5) and $\{p_{1,j}(r_1,\ldots,r_{k-1}), p_{2,j}(r_1,\ldots,r_{k-1}), g_j(r_1,\ldots,r_{k-1})\}_{j\in\mathcal{C}}$, one can compute (4). So we can safely exclude (4) from the view as it does not add additional information. Furthermore, because of the structure of g, one can compute $g(r_1,\ldots,r_{k-1},x_k)$ just by using $h_{k-1}(x_{k-1})$, $K_{i,r}$, for $i\in[2,d]$ and $p_1(r_1,\ldots,r_{k-1},x_k), p_2(r_1,\ldots,r_{k-1},x_k)$ (see Lemma 6). Consequently, we can exclude (5) as they can be locally computed by the adversary. So, the actual view of the adversary is $\text{View}_{\mathcal{R},\mathcal{A}} = \{\{p_{1,j},\ p_{2,j},\ g_j\}_{j\in\mathcal{C}}, \{K_{i,r}\}_{i\in[d]}, \{h_i(x_i)\}_{i\in[k-1]}\}$.

Next, we will construct a simulator that produces a view as described above with distribution identical to that of the real world.

1. Sample $K^*_{i,r}$ uniformly at random for all $i\in[d]$ and the polynomials $\{h^*_i(x_i)\}$ in such a way so that they produce a valid proof. This is done by sampling $h^*_i(\omega_{j,i})$ for all $j\in[2,2d_i]$ uniformly at random and setting $h^*_i(\omega_{1,i}) = K^{(i)} - \sum_{j\in[2,2d_i]} h^*_i(\omega_{j,i})$ where $K^{(i)} = h^{*(i-1)}(r_{i-1})$ when $i>1$ and $K^{(1)} = K + aK_r$.
2. Sample $\{p^*_{1,j}, p^*_{2,j}, g^*_j\}_{j\in\mathcal{C}}$ uniformly at random and output: $\{\{p^*_{1,j}, p^*_{2,j}, g^*_j\}_{j\in\mathcal{C}}, \{K^*_{i,r}\}_{i\in[d]}, \{h^*_i(x_i)\}_{i\in[k-1]}\}$

Clearly, the view sampled by the simulator is uniformly distributed across all accepting sumcheck proofs. So it remains to show that $\text{View}_{\mathcal{R},\mathcal{A}}$ follows the same distribution with probability $1-\text{negl}(\lambda)$ (over the verifier's challenges). This is proven by the following Lemma:

Lemma 6. *For any $p^*_1, p^*_2 : \mathbb{F}^k \to \mathbb{F}$ such that $K = \sum p^*_1(\omega)p^*_2(\omega)$ and $p^*_i(r_1,\ldots, r_{k-1},x_k) = p_i(r_1,\ldots,r_{k-1},x_k)$ where $i=1,2$, with probability $1-\frac{1+\sum 2d_i}{|\mathbb{F}|}$ there exist unique polynomials (padded with randomness) $\hat{p}_1,\hat{p}_2$ that produce $\{\{p_{1,j},p_{2,j}\}_{j\in\mathcal{C}}$ and a unique masking polynomial g that produces $\{\{K_{i,r}\}_{i\in[d]}, \{h_i(x_i)\}_{i\in[k-1]}\}$ and $\{g_j\}_{j\in\mathcal{C}}$.*

Proof. First, note that because $|\mathcal{C}|\le t$, then due to the security guarantees of the packed secret sharing there will always exist padded by randomness polynomials of p^*_1, p^*_2 that produce the secret shares $\{\{p_{1,j},p_{2,j}\}_{j\in\mathcal{C}}$. Likewise, we will argue about g, but first we will show how to compute it using $\{\{K_{i,r}\}_{i\in[d]}, \{h_i(x_i)\}_{i\in[k-1]}\}$.

Recall that $g(x_1,\ldots,x_k)$ is constructed such that all its evaluations on $\mathbb{H}_1\times\ldots\times\mathbb{H}_k$ are zero except from the points $(\omega_{1,1},\ldots,\omega_{i-1,1},\mathbb{H}_i,\omega_{i+1,1},\ldots,\omega_{k,1})$ where g is randomly sampled. In practice, g has the following form:

$$g(x_1,\ldots,x_k) = \sum_{i\in[k]} g^{(i)}(x_i)\cdot\prod_{j\in[k]\setminus i} L_{j,0}(x_j)$$

where $g^{(i)}(\omega_{i,1}) = 0$ for all i except from $i=k$. Based on this structure, observe that the sumcheck polynomial $h_{g,i}$ of the i-th round produced by g has the following form:

$$\sum_{\mathbf{w}\in\mathbb{H}_{i+1}\times\ldots\mathbb{H}_k} g(r_1,\ldots,r_{i-1},x_i,\mathbf{w}) = l_i g^{(i)}(x_i) + L_{i,0}(x_i)(K^{(i)}_r - l_i\sum_{\omega\in\mathbb{H}_i} g^{(i)}(\omega))$$

Where $l_i = \prod_{j\in[i-1]} L_{1,i}(r_i)$ and $K_r^{(i)} = \sum_{\mathbf{w}\in\mathbb{H}_i\times\ldots\mathbb{H}_k} g(r_1,\ldots,r_{i-1},\mathbf{w})$. Next, observe that $h_{g,i}(x_i) = a^{-1}(h_i(x_i) - \sum p_1^{(i)}(x_i,\mathbf{w})p_2^{(i)}(x_i,\mathbf{w}))$. Consequently, given $h_{g,i}(x_i)$ one can find $g^{(i)}(x_i)$ by first computing $K_r^{(i)} = \sum_{\omega\in\mathbb{H}_1} h_{g,i}(\omega)$ and then setting $g^{(i)}(\omega_{j,i}) = l_i^{-1}(h_{g,i}(\omega_{j,i}) - L_{i,0}(x_i)(K_r^{(i)} - l_i\sum_{\omega\in\mathbb{H}_i} g^{(i)}(\omega)) = l_i^{-1}h_{g,i}(\omega_{j,i})$ for all $j\in[2,2d_i]$. Finally, one can compute $g^{(k)}$ by setting $g^{(k)}(\omega_{j,k}) = K_{i,r}$ for all $j\in[2,d]$ and $g^{(k)}(\omega_{1,k}) = K_{1,r} - \sum_{i\in[2,k]}\sum_{\omega\in\mathbb{H}_i} g^{(i)}(\omega)$.

Note however that the above procedure does not work for any value of $a, r_1,\ldots,r_{k-1}$. In particular, it must hold that $a\neq 0$ and $r_i\notin\mathbb{H}_i'$. However the probability of sampling these points is at most $\frac{1+2\sum_{i\in[k-1]} d_i}{|\mathbb{F}|}$ and thus negligible for large enough $\mathbb{F}$.

Finally, because of the security guarantees of packed secret sharing, and from the fact that $|\mathcal{C}|\leq t$ one can always compute $\{g_j\}_{j\in\mathcal{C}}$ from g. □

5.3 Proof of t-Zero Knowledge of our coPIOP

Here we will show that **Protocol 4** satisfies t-zero-knowledge. For this we follow the same proof as in [12], but adapt it to support our new coSumcheck protocol. In more detail, we construct a PPT simulator that with overwhelming probability it produces a view the distribution of which is identical to the view of the adversary when running the protocol in the real world.

First, observe that the real-world view of the adversary corrupting the provers with indexes $\mathcal{C}\in[N]^t$ consists of the following: (1) the secret shared polynomials $\{p_j, p_{L,j}, p_{R,j}, p_{O,j}, p_{1,j}, p_{2,j}, p_{3,j}\}_{j\in\mathcal{C}}$ (2) the polynomials $p_O^{(k_2-1)}(x_{k_2})$, $p_L^{(k_2-1)}(x_{k_2})$, $p_R^{(k_2-1)}(x_{k_2})$, $p_i^{(k_1-1)}(x_{k_1})$, for $i\in[3]$ and $p_w^{(k_1-1)}(x_{k_1})$ as well as their secret shares computed by $\mathcal{C}$ and finally (3) the sumcheck proofs $\{K_{1,i,r}\}_{i\in[d]}$, $\{h_{1,i}(x_i)\}_{i\in[k_1-1]}$ and $\{K_{2,i,r}\}_{i\in[d]}$, $\{h_{2,i}(x_i)\}_{i\in[k_2]}$. Note that we omit the evaluation claims as they are derived by (2).

Based on this view, we first construct a simulator that produces (1),(2),(3) following a uniform distribution over any accepting proof and finally show that the real view actually follows a uniform distribution (with overwhelming probability with respect to the verifier's challenges). First, we construct the simulator that works as follows:

1. Sample K' uniformly at random.
2. Sample $p_Z(r_1',\ldots,r_{k_2-1}',x_{k_2})$ uniformly at random for $Z\in\{L,R,O\}$ and invoke the coSumcheck Simulator (steps 1,2) to output the first Sumcheck proof π_1.
3. Sample $p_w(r_1'',\ldots,r_{k_2-1}'',x_{k_1})$ and $p_i(r_1'',\ldots,r_{k_2-1}'',x_{k_1})$, $i\in\{1,2,3\}$ uniformly at random and invoke the coSumcheck Simulator (steps 1,2) to output the second Sumcheck proof π_2.
4. Interact with the corrupted adversaries sending the desired simulated data produced on the previous steps.

It is not hard to see that the view produced by the simulator follows a uniform distribution. What remains, is to show that the real view also follows the same distribution. For this, it is enough to show that given accepting $\mathbf{W}, \mathbf{U}_L, \mathbf{U}_R, \mathbf{U}_O$, and any accepting view, one can find $\mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3$ and $\mathbf{r}$ as well as the randomness used to produce the secret shares.

Finding $\mathbf{r}$. Note that in order to find $\mathbf{r}$ observe that for any $\omega_{i,k_1} \in \mathbb{H}'_{k_1}$ $p_w^{(k_1-1)}(\omega_{i,k}) = p_w(r''_1, \ldots, r''_{k_1-1}, \omega_{i,k_1}) = p_{\mathbf{W}_i^*}(r''_1, \ldots, r''_{k_1-1})$ where $p_{\mathbf{W}_i^*}$ the polynomial that interpolates the i-th row of $\mathbf{W}^*$. Note that one can directly find $\mathbf{r}[i]$ since we know all elements of the i-th row of $\mathbf{W}^*$ except from one corresponding to $\mathbf{r}[i]$. This is possible when $\beta_L(\mathbf{r}'', \mathbf{n/d+1})$ is not zero which happens with probability $\frac{\sum d_{1,i}}{|\mathbb{F}|}$.

Finding $\mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3$. For this task, we follow similar techniques as in [12]. Namely, given K'_i for $i \in [d]$ (where $K_i = \sum_j \beta((\mathbf{i}, \mathbf{j}), \mathbf{r})(\mathbf{R}_1[i,j]\mathbf{R}_2[i,j] - \mathbf{R}_3[i,j])$ so $K' = \sum K'_i$), $p_Z^{(k_2-1)}(\omega_i)$ for $Z \in [L, R, O]$ and $p_i^{(k_1-1)}(\omega_i)$, for all $\omega_i \in \mathbb{H}'_{k_1} (= \mathbb{H}_{k_2})$ one can derive two linear equations of each row i of $\mathbf{R}_1, \mathbf{R}_2$ and by solving them it produces three linear equations of $\mathbf{R}_3$. We will show that the systems of these equations are solvable with overwhelming probability. We first focus on a single row of $\mathbf{R}_1$ ($\mathbf{R}_2$ has an identical system). In particular, we want to solve a system of the form:

1. $v_{1,i} = p_{1,i}(\mathbf{r}')\mathbf{R}_1[i,0] + p_{2,i}(\mathbf{r}')\mathbf{R}_1[i,1]$.
2. $v_{2,i} = p_{3,i}(\mathbf{r}'')\mathbf{R}_1[i,0] + p_{4,i}(\mathbf{r}'')\mathbf{R}_1[i,1]$.

Where $p_{1,i}, p_{2,i}$ and $p_{3,i}, p_{4,i}$ correspond to monomials of p_L and p_1 respectively evaluated at $\mathbf{r}'$ and $\mathbf{r}''$. Note that the above system is solvable if its determinant is non-zero, namely, $p_{1,i}(\mathbf{r}')p_{4,i}(\mathbf{r}'') - p_{2,i}(\mathbf{r}')p_{3,i}(\mathbf{r}'') \neq 0$. In practice, $p_{1,i}(x_1)p_{4,i}(x_2) \neq p_{2,i}(x_1)p_{3,i}(x_2)$ since $p_{2,i}(x_1)p_{3,i}(x_2)$ has at least one different root as $p_{1,i}(x_1) \neq p_{2,i}(x_1)$, so the probability the determinant is zero with respect to $(\mathbf{r}', \mathbf{r}'')$ is $\frac{\sum d_{1,i} + \sum d_{2,i}}{|\mathbb{F}|}$.

In the same way, we argue about $\mathbf{R}_3$.

6 Functionalities Description

6.1 Functionality $\Pi^{\mathbf{PSS,ip}}$

This functionality takes as input the packed secret shares of a vector $\mathbf{v} \in \mathbb{F}^d$ and a public vector $\mathbf{b} \in \mathbb{F}^d$ and outputs $K = \sum_{i \in [d]} \mathbf{v}[i]\mathbf{b}[i]$. In particular, that protocol is partitioned in two phases:

Pre-Processing Phase. In this phase, the parties invoke Π^{Rand} and each $\mathcal{P}_i$ receives the packed secret share $[\![\mathbf{a}]\!]_i$ of a random vector $\mathbf{a} \in \mathbb{F}$. Furthermore, parties invoke Π^{PSS2SS} functionality which takes as input all shares of $\mathbf{a}$ and for each party $\mathcal{P}_i$ it returns the secret share of each element of $\mathbf{a}$, that is, $[\mathbf{a}[j]]_i$ for all $j \in [d]$.

Online Phase. Each party $\mathcal{P}_i$ computes: (1) $[\![\mathbf{v}]\!]_i + [\![\mathbf{a}]\!]_i = [\![\mathbf{v}+\mathbf{a}]\!]_i$ = and $\sum_{j\in[d]} \mathbf{b}[j] \cdot [\mathbf{a}[j]]_i = [K_a]_i$. Next, it sends both sums to $\mathcal{P}_1$ which unpacks all shares into $\mathbf{a}+\mathbf{v}$ and K_a. Finally, $\mathcal{P}_1$ sends $K = \sum_{j\in[d]} \mathbf{b}[j](\mathbf{v}[j]+\mathbf{a}[j]) - K_a$ to all other parties.

Security Analysis. Note that $\Pi^{\text{PSS,ip}}$ is secure against an semi-honest adversary controlling up to $t \leq N-d$ corruptions. First, let $\text{View}_{\mathcal{R},\mathcal{A}}(K)$ be the real-world view of the adversary. This consists of the following: $\mathbf{v}' = \mathbf{v}+\mathbf{a}$, $[\![\mathbf{v}]\!]_i, [\![\mathbf{a}]\!]_i, \{[\mathbf{a}[j]]_i\}_{j\in[d]}$ where $i \in \mathcal{C}$ and K_a, where $\mathcal{C} \in [N]^t$ is the index set of the corrupted parties. Note that we omit the shares sent by the honest parties as they can be derived from $\mathbf{v}+\mathbf{a}$ and K_a.

We first show that this view follows a uniform distribution. That is because for a fixed view, and for any $\mathbf{v} \in \mathbb{F}^d$ such that $K = \sum_{j\in[d]} \mathbf{b}[j]\mathbf{v}[j]$, we can find the masking randomness $\mathbf{a} \in \mathbb{F}^d$. Indeed, $\mathbf{a} = \mathbf{v}' - \mathbf{v} \in \mathbb{F}^d$ for which it must hold that $K_a = \sum_{j\in[d]} \mathbf{b}[j]\mathbf{a}[j]$. Based on this, we can construct a simulator that is given K and the secret shares held by the corrupted parties and produces a view following a uniform distribution. This is simply done by first sampling $\mathbf{v}'$ and then computing $K_a = K - \sum_{j\in[d]} \mathbf{b}[j]\mathbf{v}'[j]$. Then, using $\mathbf{v}'$ and $[\![\mathbf{v}+\mathbf{a}]\!]_i, i \in \mathcal{C}$ it computes the secret shares of the honest verifiers. Note that this is possible since $t \leq N-d$. Likewise, using K_a and $[K_a]_i, i \in \mathcal{C}$ it computes $[K_a]_i$ for honest parties.

6.1.1 Batched $\Pi^{\text{PSS,ip}}$

Here we propose an optimization of $\Pi^{\text{PSS,ip}}$ when parties wish to compute the inner product of the multiple packed values over a public vector, that is, compute $K_i = \sum_{j\in[d]} \mathbf{b}[j]\mathbf{v}_i[j]$ for $i \in [m]$. Naively, parties could invoke $\Pi^{\text{PSS,ip}}$ m times. This, however, incurs a per-party complexity of $\mathcal{O}(m+dm)$ (excluding $\mathcal{P}_1$'s steps) which mainly stems from the aggregation of Shamir's secret shares. Here, we show how to reduce this complexity to only $\mathcal{O}(m)$. This is done as follows:

Pre-Processing Phase. In this phase, the parties invoke $\Pi^{\text{Rand,Matrix}}$ and each $\mathcal{P}_i$ receives the packed secret shares $[\![\mathbf{A}[j,:]]\!]_i$ and $[\![\mathbf{A}[:,j]]\!]_i$ of a random matrix $\mathbf{A} \in \mathbb{F}^{m\times d}$ for all $j \in [m]$.

Online Phase. Each party $\mathcal{P}_i$ computes $[\![\mathbf{v}_j + \mathbf{A}[j,:]]\!]$, $j \in [m]$ and $[\![\mathbf{K}_a]\!] = \sum_{j\in[d]} \mathbf{b}[j][\![\mathbf{A}[:,j]]\!]_i$. Finally, it sends both shares to $\mathcal{P}_1$ which finds $\mathbf{v}_j + \mathbf{A}[j:]$ and $\mathbf{K}_a \in \mathbb{F}^m$ and outputs $\mathbf{K} \in \mathbb{F}^m$ such that $\mathbf{K}[j] = \sum_{i\in[d]} \mathbf{b}[i](\mathbf{v}_j + \mathbf{A}[j:]) - \mathbf{K}_a[j]$.

Following the security analysis of $\Pi^{\text{PSS,ip}}$ it is not hard to see that the above protocol is secure against a semi-honest adversary controlling up to $t \leq N-d$ corruptions.

Furthermore, observe that the time complexity for each $\mathcal{P}_i$ except from $\mathcal{P}_1$, is $\mathcal{O}(d+m)$ while for $\mathcal{P}_1$ it is $\mathcal{O}(Nm)$. Note that the last one can be reduced to $\mathcal{O}(m)$ by having each each party do a part of $\mathcal{P}_1$'s work.

$\Pi^{\mathbf{PSS,ip}}$: This functionality takes as input the packed secret shares of $\mathbf{v} \in \mathbb{F}^d$, such that $d + t \leq N$ from all parties and a public $\mathbf{b} \in \mathbb{F}^d$, and outputs $K = \sum_{i \in [d]} \mathbf{v}[i]\mathbf{b}[i]$.

Pre-Processing. All parties collaborate to compute the packed secret shares of a random vector $\mathbf{a} \in \mathbb{F}^d$ and a secret share of each element of $\mathbf{a}$. Finally each $\mathcal{P}_i$ ends up with $[\![\mathbf{a}]\!]_i$ and $[\mathbf{a}[j]]_i$ for all $j \in [d]$.

Protocol. Each party holds $\mathbf{b}$, $[\![\mathbf{v}]\!]_i, [\![\mathbf{a}]\!]_i$ and $[\mathbf{a}[i]]_{i \in [d]}$.

1. $\mathcal{P}_i$: Send $[\![\mathbf{v} + \mathbf{a}]\!]_i$ and $[K_a]_i = \sum_{j \in [d]} \mathbf{b}[j][\mathbf{a}[j]]_i$ to $\mathcal{P}_1$.
2. $\mathcal{P}_1$: Compute $\mathbf{v} + \mathbf{a} \in \mathbb{F}^d$ and K_a and output $\sum_{j \in [d]} \mathbf{b}[j](\mathbf{a}[j] + \mathbf{v}[j]) - K_a$.

6.2 Functionality $\Pi^{\mathbf{Enc}}$

Our collaborative code, utilizes the Π^{Enc} functionality of [12] which essentially enables N parties holding their private inputs $\mathbf{X}_i \in \mathbb{F}^{k/N \times m}$ to collaboratively evaluate an $\text{RS}^{\otimes 2}$ codeword over their joint input $\mathbf{X} = (\mathbf{X}_1, \ldots, \mathbf{X}_N) \in \mathbb{F}^{k \times m}$. Crucially we wish the per-party complexity to be $\mathcal{O}(|\mathbf{X}|/N)$. This is done as follows:

Pre-Processing. All parties, invoke the $\mathcal{F}_{rand}$ functionality (see [7] for detailed description) with input l, which generates packed secret shares of the columns of the random matrix $\mathbf{R} \in \mathbb{F}^{k' \times l}$. Let $\mathbf{R}' \in \mathbb{F}^{N \times l}$ be the secret-shared matrix. $\mathcal{F}_{rand}$ returns $\mathbf{R}'[i, :] \in \mathbb{F}^l$ to $\mathcal{P}_i$ for each $i \in [N]$.

Protocol Π^{Enc}: Protocol for computing $\mathbf{C} = \text{Enc}_{\text{RS}^{\otimes 2}}(\mathbf{X}) \in \mathbb{F}^{N \times M}$. At the end of the protocol, each party $\mathcal{P}_i$ ends up with the i-th row of $\mathbf{C}$.

1. $\mathcal{P}_i$: Expands the matrix $\mathbf{X}_i$ to $\tilde{\mathbf{X}}_i \in \mathbb{F}^{k' \times \frac{m}{N}}$ by padding it with $k' - k$ random rows sampled uniformly at random. For each column $j \in [\frac{m}{N}]$, interpolate the k'-degree polynomial $c^{(i,j)} : \mathbb{F} \to \mathbb{F}$ such that $c^{(i,j)}(z_1^h) = \tilde{\mathbf{X}}_i[j][h]$, for all $h \in [k']$. Finally, evaluate the matrix $\mathbf{X}'_i \in \mathbb{F}^{N \times \frac{m}{N}}$ such that $\mathbf{X}'_i[h][j] = c^{(i,j)}(\omega_1^h)$, for all $j \in [\frac{m}{N}]$ and $h \in [N]$.
2. $\mathcal{P}_i$: Send the j-th row of $\mathbf{X}'_i$ to $\mathcal{P}_j$, for all $j \in [N] \backslash i$. Wait for the i-th row of $\mathbf{X}'_j$ for all $j \in [N]$.
3. $\mathcal{P}_i$: Form $\mathbf{X}'[i, :]$ by merging $\mathbf{R}'[i, :]$ with $\mathbf{X}'[i, :]$. Note that this step is the major difference between our encoding functionality and that of [12]. In particular, in [12] these two vectors are concatenated while in our functionality, they are merged in such a way as described in Section 4.1. Interpolate the $m + l$-degree polynomial $r^{(i)} : \mathbb{F} \to \mathbb{F}$ such that $r^{(i)}(z_2^j) = \mathbf{X}'[i, j]$ and compute $\mathbf{C}[i, :] \in \mathbb{F}^M$, such that $\mathbf{C}[i, j] = r^{(i)}(\omega_2^j)$, for all $j \in [M]$.

References

[1] G. Arnon, A. Chiesa, G. Fenzi, and E. Yogev. WHIR: reed-solomon proximity testing with super-fast verification. In *EUROCRYPT*, pages 214–243, 2025.

[2] E. Ben-Sasson, D. Carmon, Y. Ishai, S. Kopparty, and S. Saraf. Proximity gaps for reed–solomon codes. *Journal of the ACM*, 70(5):1–57, 2023.

[3] E. Ben-Sasson, A. Chiesa, and N. Spooner. Interactive oracle proofs. In *TCC*, pages 31–60, 2016.

[4] B. Bünz, G. Fenzi, R. D. Rothblum, and W. Wang. Tensorswitch: Nearly optimal polynomial commitments from tensor codes. *IACR Cryptol. ePrint Arch.*, 2025:2065, 2025.

[5] R. Canetti, Y. Chen, J. Holmgren, A. Lombardi, G. N. Rothblum, R. D. Rothblum, and D. Wichs. Fiat-shamir: from practice to theory. In *ACM STOC*, pages 1082–1090, 2019.

[6] A. Chiesa, P. Manohar, and N. Spooner. Succinct arguments in the quantum random oracle model. In *TCC*, pages 1–29, 2019.

[7] S. Garg, A. Goel, A. Jain, G. Policharla, and S. Sekar. zksaas: Zero-knowledge snarks as a service. In *USENIX Sec*, pages 4427–4444, 2023.

[8] U. Haböck and A. Kindi. A note on adding zero-knowledge to starks. *IACR Cryptol. ePrint Arch.*, 2024.

[9] J. Holmgren. On round-by-round soundness and state restoration attacks. *IACR Cryptol. ePrint Arch.*, 2019.

[10] T. Liu, X. Xie, and Y. Zhang. zkcnn: Zero knowledge proofs for convolutional neural network predictions and accuracy. In *CCS*, pages 2968–2985, 2021.

[11] C. Pappas and D. Papadopoulos. Sparrow: Space-efficient zksnark for data-parallel circuits and applications to zero-knowledge decision trees. In *ACM CCS*, pages 3110–3124, 2024.

[12] C. Pappas, D. Papadopoulos, and C. Papamanthou. Code-based scalable collaborative snarks. In *IEEE SP*, 2026.

[13] P. Valiant. Incrementally verifiable computation or proofs of knowledge imply time/space efficiency. In *TCC*, pages 1–18, 2008.